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Stated or Sampled? Post-Training Narrows a Language Model's Sampled Differential Diagnosis, Not the One It States

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Abstract

A language model's differential diagnosis can be read from the list it states in one answer or from the spread of its answers over repeated samples, the readout behind pass@k, self-consistency and sampling-based uncertainty. We compare the two readouts across base and post-trained versions of open 7–32B models from three families (Qwen, OLMo and Mistral) on DDXPlus, whose synthetic patients carry a reference differential, and on 897 published case reports. Direct-answer post-training narrows the sampled differential in every family: post-trained models give 1.3–2.4 distinct answers per 32 samples, against 3.8–4.9, and with each model prompted at its best they lose 13–30 points of pass@32 (7–18 after correcting for chance); reasoning modes narrow it less or not at all. The stated list does not narrow. For the stronger models it stays close to the base model's averaged list for the same patient, it varies little between calls, and for every Qwen model it contains the true diagnosis more often. Zero-shot, severe conditions, also from published must-not-miss lists, are named more often on average, but specific ones are lost. In four of six pairs the first-answer gain of post-training disappears once base and post-trained models get the same prompt and decode greedily. On DDXPlus, majority voting stops helping and agreement between samples weakens as an error signal; with Qwen3-8B's thinking mode on, agreement is the better signal again. On the case reports, post-trained samples likewise cover fewer of the alternatives the reports' authors considered while one stated list covers as many, and the stated probability becomes at least as good an error signal as agreement. Studies of diagnostic breadth should report which readout they use.

Data and Code Availability. All data are public: DDXPlus (Fansi Tchango et al., 2022b) (synthetic patients) Phenopacket Store v0.1.27 (Danis et al., 2025) and MedCaseReasoning (Wu et al., 2025b) (cases from published literature). All models are public checkpoints (Table 5). Prompts and an example vignette are in Appendix B; code and sampled outputs will be released with the camera-ready version.

Institutional Review Board (IRB). This work uses only public synthetic or previously published, de-identified data and involves no human subjects, so it did not require IRB review.

1. Introduction

Clinicians reason over a differential diagnosis, a ranked set of conditions consistent with the presentation, and the conditions they keep in view determine which tests are ordered and which dangerous possibilities are ruled out. A language model's differential can be read in two ways: from the list it states in one answer, or from the distribution of its answers

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over repeated samples. Much current practice uses the second. Pass@k, self-consistency voting (Wang et al., 2023) and sampling-based uncertainty (Kuhn et al., 2023; Farquhar et al., 2024) draw on repeated samples, and so does the evidence that reinforcement learning sharpens models, which win at pass@1 while their base models solve more problems given many samples (Yue et al., 2025; Wu et al., 2025a; Oh et al., 2026). The two readouts agree for a model that samples from the distribution it states. Instruction-tuned models do not: they verbalize distributions they fail to sample from (Meister et al., 2025; Zhang et al., 2026; Jang et al., 2026), and one aligned answer can cover what many base-model samples spread out (Lake et al., 2025). We ask what this means for differential diagnosis, where every case has a reference differential and a true diagnosis, and what it does to the tools that read the differential from samples. This is a replication and extension in a clinical setting, not a new phenomenon.

The design holds the pretrained weights fixed and varies only the post-training: base/instruct pairs for Qwen2.5-7B, Qwen3-8B, Qwen2.5-32B and Mistral-7B; every released stage (SFT, DPO, RL) of OLMo 3 7B (Team Olmo et al., 2025) and OLMo 3.1 32B, plus a 7B checkpoint trained with RL directly from the base; three medical models fine-tuned from Qwen2.5-7B-Instruct; and Qwen3-8B with its thinking mode on. Each final pair is also run with worked examples in the prompt, so that base models answer in the requested format. Our measurements (Appendix A maps each to its evidence and checks):

- **Sampled answers collapse** in every family: post-training costs 13–30 points of pass@32 at each model's best prompt (7–18 after subtracting what samples reach by chance), and most of its first-answer gain disappears with a shared prompt and greedy decoding (§4.1).
- **The stated list does not.** It stays close to the base model's view of the patient and varies little between calls; for the smaller Qwen models it names the true diagnosis and the reference's conditions more often, and severe conditions, also from published must-not-miss lists, are named more often on average, though specific ones are lost (§4.2).
- **Sampling-based methods lose value on DDXPlus**: voting stops helping and agreement weakens as an error signal, also on cases where the samples disagree; combining it with the stated probability helps modestly, and thinking mode restores agreement (§4.3).
- **On published case reports the split replicates against the authors' own alternatives**: post-trained samples cover fewer of them while one list covers as many, voting helps little for any model, and after post-training the stated probability ranks errors at least as well as agreement (§4.4).

2. Related work

Post-training and the sampled distribution. RL with verifiable rewards improves pass@1 while base models win at large k on math and code (Yue et al., 2025; Wu et al., 2025a); a "sharpening tax" appears across 14 base/post-trained pairs on agentic tasks (Oh

et al., 2026), though RL can expand the boundary in domains under-represented in pretraining or with prolonged training (Cheng et al., 2025; Liu et al., 2025b), and policy entropy collapses early in RL training (Cui et al., 2025). Instruction tuning and RLHF reduce output diversity (Kirk et al., 2024; West and Potts, 2025; Jiang et al., 2025; Murthy et al., 2025), more so with scale for SFT (Springer et al., 2026), and concentrate next-token distributions so that outputs barely respond to decoding (Yang et al., 2026). Where diversity collapses depends on the pipeline: at DPO in OLMo and OLMo 2 (Peeperkorn et al., 2025) and in a stage ablation by Liu (2026), at each OLMo-2 stage for stated-versus-sampled distributions (Jang et al., 2026), and set by training data across Olmo 3's Think, Instruct and RL-Zero lineages (Karouzos et al., 2026). Base models given a few in-context examples approach aligned models (Lin et al., 2023), and theory and experiments cast post-training as sharpening the base model's own distribution toward its mode (Huang et al., 2024; Agarwal et al., 2025; Karan and Du, 2025), or as distilling its majority vote (Prasad et al., 2025). Large-k pass@k over few answer options is inflated by chance (Dragoi et al., 2025), and chat templates alone can narrow outputs (Yun et al., 2025).

Stated versus sampled distributions and uncertainty. Models asked to verbalize a distribution match it better than their samples do (Meister et al., 2025; Jang et al., 2026), and they struggle to draw from target distributions directly (Gu et al., 2026; Zhao et al., 2026; Zhang et al., 2024); base models are better calibrated than their post-trained versions (Kadavath et al., 2022). Verbalized sampling keeps most of the base model's diversity through Tülu 3's checkpoints (Zhang et al., 2026), and aligned answers aggregate content that base models spread across samples on open-ended questions with conflicting perspectives (Lake et al., 2025). Our stated readout is the verbalized top-k elicitation of Tian et al. (2023). For error detection, sample consistency usually beats verbalized confidence and hybrids beat both (Xiong et al., 2024; Lyu et al., 2024), including for diagnosis (Savage et al., 2025; Ren et al., 2026), confidence-weighted voting improves self-consistency (Taubenfeld et al., 2025), and agreement cannot flag errors a model repeats consistently (Tan et al., 2025); aligned models' samples can collapse enough to leave sampling-based uncertainty without signal (Liu, 2026), and cross-model disagreement (Hamidieh et al., 2026) or the base model (Luo et al., 2025) can supply what self-consistency misses.

Medical models and differential diagnosis. LLMs have been evaluated as differential-diagnosis aids on clinicopathological cases (Kanjee et al., 2023; McDuff et al., 2025) and in conversation (Tu et al., 2025), and on Phenopacket Store, where they trail the phenotype-driven tool Exomiser (Reese et al., 2026; Chimirri et al., 2025); base models have been left out of clinical evaluations for failing to follow instructions (Hager et al., 2024), the format problem our worked examples address. Medical post-training includes verifier-guided SFT and PPO (Chen et al., 2025), rule-based RL (Liu et al., 2025a) and reasoning distillation (Huang et al., 2026); RL and reward models tend to favor confident answers (Leng et al., 2025; Damani et al., 2026; Mei et al., 2026). DDXPlus scores predicted against reference differentials as sets (Fansi Tchango et al., 2022a); H-DDx proposes hierarchical F1 (Lim et al., 2025), MEDDxAgent studies interactive diagnosis (Rose et al., 2025), and Puri et al. (2026) show on DDXPlus with Qwen3-8B that single-answer RL collapses sampled answers and train models to output several answers per completion. Relative to Jang et al. (2026), who compare stated and sampled distributions on synthetic and survey tar-

gets with two of our pairs, Puri et al. (2026) and Karouzos et al. (2026), we add a clinical reference distribution with a true diagnosis, real case reports scored against their authors' own differentials, must-not-miss conditions taken from published sources, a test of whose per-case view a post-trained list reproduces, prompt, temperature, decoding and chance controls, and the consequences for voting and error signals. Whether a higher temperature closes the gap differs by model: it restores pass@32 and agreement for Qwen2.5-7B-Instruct but not for Qwen3-8B (§4.1).

3. Setup

Tasks. **DDXPlus** (Fansi Tchango et al., 2022b): synthetic patients from a rule-based medical knowledge base, each with a true pathology and a reference differential, the engine's posterior over 49 conditions. We use 980 test patients (20 per condition) rendered by template as questionnaire-style vignettes without exam or test results (example in Appendix B). The prompt lists the 49 conditions and asks for the five most likely, with probabilities, as JSON, so this is a closed-set task. **MedCaseReasoning** (Wu et al., 2025b): 897 presentations written from open-access case reports (2005–2025), each with the report's final diagnosis and, in its reasoning notes, the alternatives the authors considered. The differential is open-ended; a judge model from a family we do not evaluate on these cases decides whether a name matches the final diagnosis or one of the authors' alternatives, and a second one gives a strict grading (Appendix B). **Phenopacket Store** (Danis et al., 2025): 1,401 published rare-disease cases (749 diseases) as phenotype vignettes, graded via Mondo (Appendix B).

Models and sampling. Qwen2.5-7B, Qwen3-8B, Qwen2.5-32B and Mistral-7B-v0.3, each with its official post-trained version; OLMo-3-1025-7B and OLMo-3-1125-32B with the SFT, DPO and final RLVR (Instruct) checkpoints of OLMo 3 7B and OLMo 3.1 32B, plus OLMo-3-7B-RL-Zero-General, trained with RL from the 7B base; and AlphaMed, HuatuoGPT-o1 and m1, fine-tuned from Qwen2.5-7B-Instruct (Table 5). We draw 32 samples per case with vLLM at T=0.7 (top-p 0.95) and, for the 7–8B models, T=1 (top-p 1), each model with its own chat template (base models without one get a post-trained sibling's) and Qwen3's thinking mode off unless stated. A sample's *top answer* is valid if it is one of the 49 conditions and its *list* if it names at least one; invalid answers count as wrong. With this zero-shot prompt base models are often invalid (up to 47% of samples for OLMo and 76% for Mistral), so every final pair is re-run with three worked examples from the training split (Appendix F.1), on which every model is valid in over 99% of samples.

Readouts. The *sampled* readout is the distribution of top answers over a case's samples: pass@k (Chen et al., 2021), distinct top answers, the reference mass they cover (coverage@32), and total-variation distance (TV) of their empirical distribution to the reference. The *stated* readout is one sample's list, with its probabilities renormalized over the conditions it names (uniform if missing), averaged per case over valid samples: whether it contains the true pathology among its top k (recall@k), its TV to the reference (also with the true pathology removed from both), the reference mass it names, its *conditional* TV (to the reference restricted to the named conditions and renormalized, a scale-free measure of how it allocates probability), DDXPlus's set F1 against the reference's support (DDF1),

and whether it names each severe condition (DDXPlus severity 1–2, the dataset's label) that the reference weights at least 0.1. The *single-call answer* is the top answer of a model's first valid sample.

Statistics. DDXPlus has 20 cases per true pathology, so intervals are 95% percentile intervals from a bootstrap that resamples the 49 pathologies with all their cases (10,000 resamples) unless stated; p-values use the add-one correction. The 112 headline tests (14 metrics fixed in advance, eight zero-shot settings) are primary and Holm-corrected together; every control, stage, few-shot and per-condition analysis is secondary, corrected within its own family (Appendix J), and read as a check on them. Case-level intervals in the appendix are up to three times narrower, and some appendix tables come from an earlier scorer or a separate script, so values of the same quantity differ by up to 0.006. AUROCs use average ranks for ties. A *setting* is one model pair at one temperature and prompt. With five to six pipelines, results describe these checkpoints rather than post-training in general.

4. Results

4.1. Post-training collapses the sampled answers

Table 1 compares each base model with its post-trained versions. Post-trained models give 1.3–2.4 distinct top answers per 32 samples at T=0.7, against 3.8–4.9. At 7–8B the post-trained Qwen model is far better on its first answer (+12.6 to +27.5 points), but its pass@k curve crosses the base model's at k≈4–8, and at k=32 the base model leads by 9.7–14.6 points (Figure 2); at 32B the zero-shot losses are small and not significant. Because base models often answer the zero-shot prompt invalidly, we also compare each base model given worked examples with its post-trained model at its better prompt: post-training then costs 13.2–29.9 points of pass@32 in all six final pairs, including both 32B pairs and Mistral, also with example lists taken from another model instead of the reference (Appendices F.1–F.8). Part of pass@32 is chance: scored against another patient's diagnosis, base models' samples still reach 0.05–0.19 and post-trained ones 0.01–0.04, and after subtracting these the loss with worked examples is 6.7–17.8 points (Appendix F.5). Pooled across pairs, the zero-shot loss is 11.9 points [8.5, 15.4] (Table 35), and it holds on the cases where both models give at least 16 valid answers (Appendix F.7).

Two controls separate sharper sampling and prompt format from lost breadth. Sampled hotter, Qwen2.5-7B-Instruct at T=1.3 matches its base model's pass@32 at T=0.7, but Qwen3-8B stays below it even at T=1.6 (Appendix F.2). And most of the first-answer gain is prompt format and sampling noise: with the same worked examples the sampled pass@1 gain falls to 1.6–5.2 points in four of six pairs, and decoded greedily, post-training significantly raises top-1 only for Qwen3-8B and Mistral, the two pairs whose few-shot gain stays large (Appendix F.3). Which stage narrows the samples depends on the recipe: RLVR in OLMo 3 7B (19.1 points of pass@32), SFT in OLMo 3.1 32B (12.1), while RL-Zero from the 7B base loses none detectably at T=1 (Appendices D.4, D.5). Qwen3-8B with thinking on reaches at least its base model's pass@32 (0.815 against 0.732), though its samples stay narrow (1.76 distinct answers; Appendix G.1).

Comparison	Model	Sampled top answers pass@1	pass@32	Distinct	Cov@32	Drawn from stated lists pass@32	Distinct
Qwen3-8B, T=0.7	base	0.388	**0.732**	**3.77**	**0.323**	0.780	7.79
	post-trained	**0.559**	0.607	1.27	0.217	**0.811**	4.96
Qwen3-8B, T=1	base	0.282	**0.789**	**6.60**	**0.407**	**0.832**	9.82
	post-trained	**0.556**	0.643	1.53	0.236	0.820	5.40
Qwen2.5-7B, T=0.7	base	0.352	**0.695**	**4.91**	**0.335**	0.772	10.68
	instruct	**0.478**	0.598	1.82	0.239	**0.802**	6.81
Qwen2.5-7B, T=1	base	0.239	**0.766**	**8.23**	**0.422**	0.785	12.44
	instruct	**0.464**	0.663	2.65	0.290	**0.827**	7.76
Qwen2.5-32B, T=0.7	base	0.500	**0.795**	**3.78**	**0.352**	0.850	8.79
	instruct	**0.604**	0.746	1.68	0.276	**0.880**	6.49
OLMo 3 7B, T=0.7	base	0.178	**0.705**	**4.93**	**0.351**	**0.719**	8.66
	SFT+DPO+RLVR	0.265	0.422	2.37	0.195	0.577	5.65
	RL-Zero (from base)	**0.295**	0.572	4.06	0.286	0.701	9.66
OLMo 3.1 32B, T=0.7	base	0.326	**0.796**	**4.32**	**0.347**	0.839	8.11
	SFT+DPO+RLVR	**0.560**	0.756	2.15	0.283	**0.853**	6.75
Chance	uniform guesser	0.020	0.48	23.6	0.47		
	reference-prior guesser	—	0.44	21.9	0.56		

Table 1: Sampled readout on DDXPlus (980 cases, 32 samples). Distinct: distinct top answers per case; Cov@32: reference mass they cover. *Drawn from stated lists*: each sample's top answer replaced by a draw from its own stated list. Bold: better model of each pair (OLMo: of the block; the SFT and DPO stages are in Tables 9 and 10). Invalid answers count as wrong, which mostly penalizes base models. Chance: guessers that ignore the patient, drawing uniformly from the 49 conditions or from the reference differential averaged over test cases.

4.2. The stated list stays close to the base model's view and varies little

For each patient we compare each post-trained list with the mean of its base model's lists for the same patient (Table 27). In all 15 settings it is closer to that mean than to the base model's means for other patients with the same diagnosis (TV −0.028 to −0.076), and closer than to other families' base models (by 0.03–0.14). Against every other model's mean for the same patient (Table 3), the lists of Qwen3-8B, Qwen2.5-32B and OLMo 3.1 32B are closest to their own base model's among base models but about as close to the nearest other post-trained model's; Qwen2.5-7B-Instruct and OLMo-3-7B-Instruct are not closest to their own base model's. Post-trained lists barely vary between calls (TV 0.10–0.33 to the mean of the model's other lists, against 0.37–0.69 for base lists). For the stronger models, then, a post-trained list is a stabilized version of the base model's view, and the stronger post-trained models converge on one another.

Whether the list improves depends on the model (Table 2). For every Qwen model and OLMo 3.1 32B, one post-trained list contains the true pathology among its top five 12–24 points more often (e.g. Qwen3-8B 0.67 → 0.80), is closer to the reference (TV −0.043 to −0.088) and allocates probability among the named conditions more like it (conditional TV −0.025 to −0.078); OLMo 3 7B's lists do not improve. The gains hold on complete lists, with the true pathology removed and at the full test split's pathology frequencies (Appendices E.2, F.6). With worked examples they persist for the smaller Qwen models

and Mistral, persist in TV but not recall for Qwen2.5-32B, and weaken for OLMo 3.1 32B; pooled across pairs recall rises in every prompt setting, and the stated list gains more for Qwen than for the other families (zero-shot recall@5 by 6.8 points [3.6, 10.0]; Table 35). Nor do the lists' probabilities concentrate, except for OLMo 3 7B (Appendix F.7). Averaging calls closes most of the gap: the mean of five base-model lists is about as close to the reference as one post-trained list (Qwen3-8B 0.699 against 0.700; diamonds in Figure 1a), and with five lists on both sides post-training is closer in only four of eleven settings (Appendix G.2).

Severe conditions. Post-trained lists name the severe conditions that the reference warrants (weight at least 0.1) more often in seven of eight zero-shot settings (+6.0 to +13.8 points, six surviving correction; OLMo 3.1 32B +2.2, n.s.), but one list still names only 0.32–0.47 of them, against 0.975 for the reference's top five, and specific conditions are lost, most often acute pulmonary edema (e.g. Qwen2.5-32B 0.63 → 0.44; Table 12), usually in favor of pulmonary embolism or acute coronary syndromes (Appendix F.6). Per case, though, post-trained lists leave out a severe true pathology 9–18 points less often and name none of the warranted severe conditions about 10 points less often (Appendix F.7). With must-not-miss conditions taken instead from diagnostic-harm data and the chest-pain guideline (Newman-Toker et al., 2024; Gulati et al., 2021), post-trained lists name 9–24 points more of them in seven of eight zero-shot settings, and only OLMo, Mistral and the few-shot Qwen2.5 models lose specific ones (Appendix F.4). Asked explicitly for up to three dangerous conditions to rule out, post-trained lists name 6–15 points more of the warranted severe conditions, partly because the lists get longer (0.01–0.11 of the gain remains against two combined standard lists), at a cost of one to two more severe conditions per list that the reference does not warrant (Table 21). These vignettes lack the ECG and biomarkers that separate the coronary syndromes, so these shifts track the reference engine rather than clinical correctness.

4.3. Consequences for sampling-based methods

Aggregation. Majority voting over 32 samples raises base models' accuracy by 3.4–8.4 points at T=0.7 but adds at most 1.9 for the final post-trained models (Table 29); models trained to reason before answering (HuatuoGPT-o1, m1) and a chain-of-thought prompt still gain 3.8–4.8 (thinking mode +1.5, n.s.), and so does Qwen2.5-7B-Instruct at T=1.3 (+5.6). The top three of one post-trained Qwen list contains the true pathology 5.6–13.7 points more often than the three most frequent of 32 samples, but only because collapsed samples offer fewer than three answers: matched for size, the list is no better than its samples (Appendix F.6).

Uncertainty. We score each signal by how well it ranks the correctness of the single-call answer (AUROC); the stated probability needs no extra calls, *agreement* (the share of 31 further samples with the same top answer) needs them. For base models agreement is the better signal by 0.07–0.26. Its AUROC falls after post-training in every pair (e.g. Qwen3-8B 0.810 → 0.655; Figure 1b), individually significantly only for Qwen, and by 0.10 [0.06, 0.12] pooled, consistent with Liu (2026). This is not only ties: where neither model's samples are unanimous, agreement's AUROC still falls in every pair, and accuracy at 50% coverage shows the same reversal (Appendix F.7). Which signal is better then

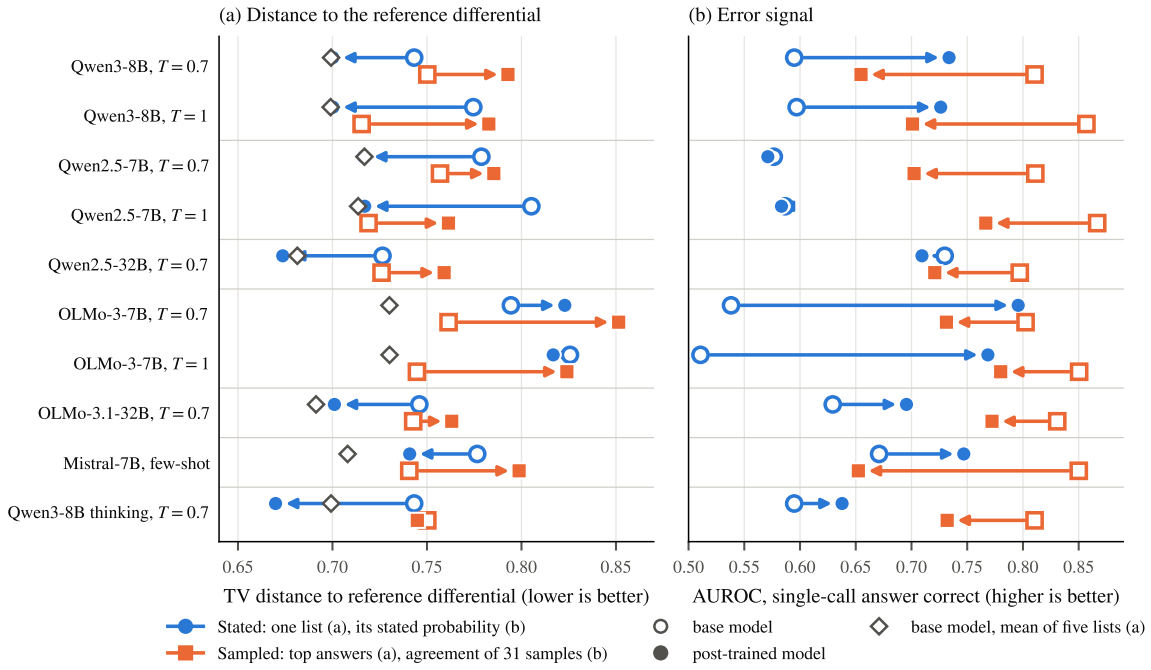


Figure 1: Base (hollow) → post-trained (filled) on DDXPlus (980 cases; OLMo: base → final Instruct; Mistral-7B with the few-shot prompt; the last row compares Qwen3-8B-Base with Qwen3-8B with thinking on). (a) TV to the reference differential of one stated list (blue circles) and of the sampled top answers (orange squares); grey diamonds: the mean of the base model's first five lists. (b) AUROC for the single-call answer being correct, of its stated probability (blue) and of agreement across 31 further samples (orange).

depends on model and prompt: zero-shot, the stated probability wins for Qwen3-8B (+0.079 [0.011, 0.142]) and agreement for Qwen2.5-7B-Instruct and OLMo-3.1-32B-Instruct, and with worked examples the stated probability wins for Qwen3-8B, Qwen2.5-7B-Instruct and Mistral-Instruct. Thinking returns Qwen3-8B to the base pattern (agreement better by 0.094 [0.027, 0.170]), and so does sampling Qwen2.5-7B-Instruct at T=1.3. Across 56 model–setting combinations evaluated on held-out cases, a logistic combination of the two signals fitted on 100 labeled cases adds only 0.030 [0.013, 0.044] AUROC over agreement for post-trained models (Appendix G.3).

4.4. Real cases and medical post-training

Published case reports. On 897 presentations written from published case reports, scored against each report's final diagnosis and against the alternatives its authors considered (Table 32), the readout split replicates. After post-training the 32 samples cover 8–14 points fewer of the authors' alternatives for the three Qwen pairs (OLMo 3 7B −3.0, n.s.), while one stated list covers them as before (within 1.4 points) and contains the final diagno-

Comparison (T=0.7)	Model	TV to reference↓ One list	TV to reference↓ Samples	TV to reference↓ List w/o gold	Reference mass named	Cond. TV↓	DDF1	Recall @5	Severe named
Qwen3-8B	base	0.743	**0.750**	0.833	0.334	0.492	0.276	0.666	0.347
	post-trained	**0.700**	0.793	**0.815**	**0.396**	**0.431**	**0.318**	**0.802**	**0.437**
Qwen2.5-7B	base	0.779	**0.757**	0.856	0.291	0.553	0.245	0.573	0.345
	instruct	**0.717**	0.785	**0.816**	**0.364**	**0.474**	**0.307**	**0.735**	**0.439**
Qwen2.5-32B	base	0.726	**0.726**	0.830	0.351	0.473	0.282	0.721	0.382
	instruct	**0.674**	0.759	**0.787**	**0.418**	**0.437**	**0.333**	**0.840**	**0.472**
OLMo 3 7B	base	**0.794**	**0.761**	**0.850**	**0.267**	**0.568**	**0.225**	0.470	0.256
	Instruct (RLVR)	0.823	0.851	0.884	0.257	0.604	0.220	**0.485**	**0.317**
OLMo 3.1 32B	base	0.746	**0.743**	0.835	0.334	0.497	0.273	0.652	0.370
	Instruct (RLVR)	**0.701**	0.763	**0.812**	**0.391**	**0.467**	**0.312**	**0.779**	**0.392**
Chance	patient-blind	0.794	0.774	—	0.238	—	0.272	0.102	0.149
Oracle	reference	0.265	0.18–0.23	—	0.735	0.000	—	0.969	0.975
	one answer: reference mode	—	0.694	—	—	—	—	—	—
	one answer: true pathology	—	0.714	—	—	—	—	—	—

Table 2: Stated vs. sampled readouts on DDXPlus (T=0.7, 980 cases, valid samples). *One list*: one sample's stated differential (per-case mean over valid samples). *Samples*: empirical distribution of top answers. *List w/o gold*: TV after removing the true pathology from list and reference. *Cond. TV*: TV between the list and the reference restricted to the named conditions and renormalized (how the list allocates probability, independent of how much mass it names). DDF1: set F1 of the named conditions against the reference's support. *Recall@5*: the list contains the true pathology. *Severe named*: share of severe conditions with reference probability ≥ 0.1 that one list names. Chance: the top five of the average reference differential as a fixed list, and 32 draws from that average as samples. Oracle: the reference's own top five as the list and as many reference draws as valid samples; and samplers that always give one answer, each case's reference mode or its true pathology. Bold: better model of each pair.

Post-trained model of	TV to own base mean	Closest other base model	own − it	Closest other post-trained model	own − it	Own − mean of other post-trained models
Qwen3-8B	0.368	OLMo-3-1125-32B	−0.058	Qwen2.5-32B-Instruct	+0.000†	−0.067
Qwen2.5-7B	0.464	Qwen3-8B-Base	+0.014†	Qwen2.5-32B-Instruct	+0.028	−0.041
Qwen2.5-32B	0.360	Qwen3-8B-Base	−0.036	OLMo-3.1-32B-Instruct	−0.018†	−0.090
OLMo 3 7B	0.628	Qwen3-8B-Base	+0.031	Qwen3-8B	+0.041	+0.002†
OLMo 3.1 32B	0.386	Qwen2.5-32B	−0.051	Qwen2.5-32B-Instruct	−0.006†	−0.079

Table 3: A stricter packing test (DDXPlus, zero-shot, T=0.7, 980 cases). For each post-trained model, the TV between its lists and the mean list of each other final model (five base, five post-trained) for the same patient. Negative differences: the post-trained list is closer to its own base model's view than to the comparison. †: the condition-clustered 95% interval includes zero.

sis 2–6 points more often; for Qwen3-8B and Qwen2.5-32B one post-trained list covers more of the authors' alternatives than all 32 samples together (+11.0 and +7.5 points). pass@32 for the final diagnosis falls significantly only for Qwen3-8B (−12.5). The consequences point the same way as on DDXPlus, though they are smaller: on these open-ended cases voting helps little for any model (+1.6 to +2.7 points for base models, −0.7 to +3.0 after

post-training), and after post-training the stated probability ranks correctness at least as well as agreement (significantly better for Qwen3-8B and Qwen2.5-32B, +0.085 and +0.062 AUROC), whereas for base models agreement is as good or, for OLMo, better. The two judges agree on 97.6% of names (κ 0.77) and strict grading leaves every direction unchanged (Appendix H.3).

Rare disease and medical models. On Phenopacket Store's rare-disease cases models are near the floor (pass@32 2.5–7.0%), but voting and agreement shift as on DDXPlus (Appendix H.4). Medical post-training on Qwen2.5-7B-Instruct does not narrow the samples further, and the reasoning-trained HuatuoGPT-o1 and m1 widen them (Appendix H.1); on rare disease, the RL-trained medical models state 0.67–0.69 confidence at about 2% accuracy (Appendix H.2).

5. Discussion and limitations

Jang et al. (2026) show that instruction-tuned models cannot sample from distributions they can describe. In differential diagnosis we find the same split: for the stronger models a post-trained list stays close to the base model's view of the patient and varies far less between calls, and the sampled answers collapse. Evaluations that read the differential from samples (pass@k at large k, diversity, self-consistency, agreement-based uncertainty) therefore report a loss that a clinician reading one answer would not see, and error signals that work for base models and reasoning modes work less well for post-trained models answering directly. In distance to the reference, one post-trained call adds little over five averaged base calls, and with a shared prompt and greedy decoding its first-answer gain disappears in four of six pairs, as the sharpening account predicts (Huang et al., 2024). Asking for must-not-miss conditions raises their coverage, partly through longer lists, and losses of specific conditions need checking model by model. On published case reports the split holds against the alternatives the reports' authors wrote down, and after post-training the stated probability is at least as good an error signal as agreement there too. We suggest that studies of diagnostic breadth report which readout they use and whether the model reasons first, and that the error signal be chosen on a small labeled sample from the target population.

Limitations. DDXPlus is synthetic and closed-set: its reference differential is a rule-based engine's posterior over questionnaire answers without exam or test results, sometimes clinically implausible (Appendix B), its test set is class-balanced, and the prompt lists all 49 conditions, so our metrics measure agreement with that engine, not clinical correctness, and no clinician reviewed lists or references. Results hold when reweighted to the full test split's pathology frequencies (Appendix F.6). Severity is the dataset's label or a published list mapped onto its 49 conditions, and the prompt asks for the most likely conditions, not must-not-miss ones. The few-shot controls use one set of example patients, and the temperature controls cover the 7–8B models only. We study open 7–32B models from three families, one reasoning mode and no frontier models. The case-report judges and the extraction of the authors' alternatives are language models not validated against clinicians, and published cases may appear in pretraining data; Phenopacket accuracy is near the floor.

Statement on AI Use. An AI assistant (Claude, Anthropic) was used to write and run analysis code, to draft and edit text, and to check citations. The authors designed the study, reviewed all code, results and text, and take responsibility for the content.

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Appendix A. Claims and the evidence for them

Table 4 lists each claim with its primary evidence, the checks under which it holds and those under which it weakens or fails. Primary evidence comes from the 112 headline tests (§3); everything else is a secondary check, and Table 14 gives the same information control by control.

Claim	Primary evidence	Holds under	Weaker or fails under
(a) A post-trained list stays close to the base model's view of each patient and varies little between calls (§4.2)	A post-trained list is closer to its own base model's mean list for the same patient than to the means for other patients with the same diagnosis, in all 15 settings (TV −0.028 to −0.076); post-trained lists vary 0.10–0.33 between calls, base lists 0.37–0.69 (Table 27)	worked examples; thinking mode; against other families' base models (−0.03 to −0.14); for three of five pairs, against every other base model (Table 3)	about as close to the nearest other post-trained model; Qwen2.5-7B-Instruct and OLMo-3-7B-Instruct not closest to their own base
(b) Sampled answers collapse (§4.1)	pass@32 −9.7 to −14.6 points for the 7–8B Qwen pairs and −28.3 for OLMo 3 7B zero-shot; 1.3–2.4 distinct answers against 3.8–4.9 (Table 1)	worked examples of either kind (13–30 points in all six pairs); Mistral; OLMo stages; 32B with worked examples	32B zero-shot (n.s.); Qwen2.5-7B-Instruct at T=1.3; thinking mode; RL-Zero at T=1
(c) Post-training's first-answer gain is mostly prompt format and sampling noise (§4.1)	With shared worked examples the sampled pass@1 gain is 1.6–5.2 points in four of six pairs; decoded greedily, post-training raises top-1 significantly in at most two of six pairs under either matched prompt (Table 19)	both matched pairings; both 32B pairs	Qwen3-8B (+10.7 and +13.7); Mistral with shared examples (+10.8)
(d) One stated list improves robustly only for the smaller Qwen models (§4.2)	Recall@5 +12 to +24 points, TV −0.043 to −0.088, conditional TV −0.025 to −0.078 (Table 2, Table 34)	complete lists; correct and wrong top answers; true pathology removed; worked examples for Qwen3-8B and Qwen2.5-7B; thinking mode	OLMo 3 7B (no gain); OLMo 3.1 32B and Qwen2.5-32B recall with worked examples; five averaged base lists match one post-trained list
(e) Severe conditions are named more often on average, but specific ones are lost (§4.2)	Named more often in seven of eight zero-shot settings (+6.0 to +13.8 points, six surviving correction; OLMo 3.1 32B n.s.); 2–7 of 17 conditions per pair named significantly less often (Table 12)	published must-not-miss sets (+9 to +24 points in seven of eight settings)	worked examples (mean falls for Qwen2.5-32B and OLMo 3 7B); averaged lists; with published sets only OLMo, Mistral and few-shot Qwen2.5 lose specific conditions
(f) Asking for must-not-miss conditions raises coverage, partly by lengthening the list (§4.2)	+6 to +15 points of warranted severe conditions named, 0.01–0.11 beyond two combined standard lists (Table 21)	published must-not-miss sets (+15 to +26 points; 8–20 beyond two lists)	costs one to two more unwarranted severe conditions per list
(g) Voting stops helping and agreement weakens as an error signal (§4.3)	Voting +3.4 to +8.4 points for base models, at most +1.9 for post-trained ones; agreement's AUROC falls in every family (Table 29)	worked examples; Mistral; rare-disease cases	case reports (voting gains small for every model); models that reason first; thinking mode; Qwen2.5-7B-Instruct at T=1.3; agreement drop n.s. for OLMo
(h) The better error signal depends on model and prompt; combining helps post-trained models modestly (§4.3)	Held-out audit over 56 model–setting combinations: logistic combination +0.030 [0.013, 0.044] over 31-sample agreement for post-trained models, rank average +0.019 (n.s.) (§G.3)	worked examples (stated probability better for three post-trained models)	base models and thinking mode (agreement alone as good or better); rare-disease cases (agreement better); on case reports the stated probability is as good or better after post-training
(i) On published case reports the sampled differential narrows against the authors' own alternatives while one stated list does not (§4.4)	Samples cover 8–14 points fewer of the authors' alternatives for the Qwen pairs; one list within 1.4 points; recall of the final diagnosis +2 to +6 (Table 32)	strict grading (judges agree on 97.6%, κ 0.77)	OLMo 3 7B n.s.; pass@32 loss significant only for Qwen3-8B; OLMo-3-7B-Instruct's answers vary more

Table 4: The claims and their evidence. Ranges are post-trained minus base; "n.s.": interval includes zero or test does not survive correction.

Appendix B. Prompts, vignettes and grading

DDXPlus. Each patient's coded evidence is rendered by a fixed template into a vignette of the patient's answers and relevant history; we fix known mistranslations in the English evidence descriptions. An example (test case with true pathology chronic rhinosinusitis; the reference differential gives viral pharyngitis 0.17, chronic rhinosinusitis 0.16, cluster headache 0.15, acute rhinosinusitis 0.14, bronchitis 0.14, URTI 0.12; cluster headache is implausible for this patient, a reminder that the reference is an engine's posterior rather than a clinician's differential):

> `A 13-year-old female presents for evaluation. Symptoms (patient answers): Do you have pain somewhere, related to your reason for consulting? Yes. Characterize your pain: burning. Do you feel pain somewhere? mouth, forehead, cheek(R), cheek(L), nose. How intense is the pain? (scale 0-10) 3. […] Have you lost your sense of smell? Yes. Do you have nasal congestion or a clear runny nose? Yes. Do you have a cough? Yes. Relevant history and risk factors: Have you had a cold in the last 2 weeks? Yes. […] Do you have polyps in your nose? Yes. […]`

System prompt: `You are an experienced emergency and primary care physician.` The user turn is the vignette followed by: `Based on the patient's answers above, give your differential diagnosis. Choose ONLY from this list of possible conditions: [the 49 conditions, one per line]. List the 5 most likely conditions from the list, most likely first, with your probability that each is the correct diagnosis. Use the condition names exactly as written in the list. End your answer with a JSON array in exactly this format: [{"diagnosis": "<name>", "probability": <number between 0 and 1>}, ...].` The full-distribution prompt (§E.1) instead asks for a probability on every condition, summing to one, most likely first. The plain and CoT variants are described in §D.3. Base models receive the same chat-formatted prompt; OLMo base models use their SFT checkpoint's chat template.

Phenopacket Store. The vignette lists the case's present and (up to ten) excluded HPO phenotypes with sex and age. System prompt: `You are an expert clinical geneticist and diagnostician.` Instructions: `List the 5 most likely specific diagnoses (use the specific disease or subtype name, e.g. 'Spastic paraplegia 7' rather than 'hereditary spastic paraplegia', when you can), most likely first, with your probability that each is the correct diagnosis`, followed by the same JSON format.

Published case reports. MedCaseReasoning (Wu et al., 2025b) turns open-access PMC case reports (2005–2025) into a presentation, written up to the point where the authors give a differential, and the report's final diagnosis; we use its 897 test cases. The vignette is the presentation as released. System prompt: `You are an experienced physician and diagnostician.` Instructions: `Based on the case above, give your differential diagnosis. List the 5 most likely specific diagnoses, most likely first, with your probability that each is the correct diagnosis. End your answer with a JSON array in exactly this format: [{"diagnosis":`

`"<name>", "probability": <number between 0 and 1>}, ...]`. Every distinct (case, name) pair among all samples of the eight models, after lower-casing and stripping punctuation, is graded once by Mistral-Small-24B-Instruct-2501 at temperature 0 (the Mistral family is evaluated only on DDXPlus, so no judge grades its own family here): `Does the candidate name the same disease as the final diagnosis? Count it as the same if it is a synonym, an abbreviation, or a more specific form of the final diagnosis. Count it as different if it is a broader category, a related but distinct disease, or only a complication, cause or finding. Answer with one word: yes or no.` Phi-4, from a family we do not evaluate, grades the same pairs. The first judge's grading is primary, fixed before the results; the *strict* grading, which counts a name as correct only if both judges do, is a sensitivity analysis. Neither judge was validated against clinicians; in a smoke test the first was slightly lenient (it accepted "Meningioma" for "Sclerosing meningioma").

Grading open-ended names. Each predicted name is mapped to Mondo by exact lookup against Mondo labels and synonyms (also trying any parenthesized abbreviation and the name without it), then by fuzzy matching (token-sort ratio ≥ 92). It is *exact* if it maps to the Mondo class of the case's OMIM diagnosis, and *related* if it maps to a specific ancestor group of it (at most two is-a steps up, with at most 150 descendants) or to one of its subtypes; siblings sharing only a broad parent are wrong. The grader passes hand-written checks (e.g. "Hereditary spastic paraplegia type 7 (SPG7)" is exact for SPG7, "Hereditary spastic paraplegia" is related, and Turner syndrome is wrong for Jacobsen syndrome), but we did not validate it against clinician grading, so errors at the scale of the 1–2 point effects we report on this dataset are possible.

Appendix C. Models

Appendix D. The sampled differential in detail

D.1. General post-training trades coverage for top-1 accuracy

Table 6 and Figure 2 compare each base model with its post-trained counterpart on DDXPlus. In both families the post-trained model is much better on its first answer: +27.5 points for Qwen3 and +22.5 for Qwen2.5 at T=1 (intervals $[+25.3, +29.6]$ and $[+20.6, +24.5]$). The advantage shrinks with k and reverses: the curves cross at k≈4–8, and at k=32 the base model leads by 14.6 points (Qwen3, $[+12.2, +17.1]$) and 10.3 points (Qwen2.5, $[+8.0, +12.8]$). The pattern is the same at T=0.7 for Qwen3, crossing at k=4 with a base lead of 12.4 points at k=32 ($[+10.1, +14.9]$). Unparsable base-model samples count as wrong, so these gaps are lower bounds. Qwen2.5 at T=0.7 behaves the same way: +12.6 points at k=1 $[+10.7, +14.5]$ and a base lead of 9.7 at k=32 $[+7.5, +11.9]$.

D.2. The sampled differential collapses, while top-1 calibration improves

The gap at large k reflects a collapsed *sampled* differential. Across 32 samples, the post-trained Qwen models name 1.3–2.7 distinct diagnoses per case, against 3.8–8.2 for their base models. Their answer entropy is 0.11–0.49, against 0.78–1.58 for base models and 1.87 for the reference differential (Table 6). Sampling at T=1 widens the base models'

Model	Parent	Post-training	Role
Qwen2.5-7B	n/a	none (pretrained)	base
Qwen2.5-7B-Instruct	Qwen2.5-7B	general SFT + RLHF	instruct; medical parent
AlphaMed-7B (Liu et al., 2025a)	Qwen2.5-7B-Instruct	rule-based RL	medical
HuatuoGPT-o1-7B (Chen et al., 2025)	Qwen2.5-7B-Instruct	SFT + PPO (verifier)	medical
m1-7B-23K (Huang et al., 2026)	Qwen2.5-7B-Instruct	SFT distillation	medical
Qwen3-8B-Base	n/a	none (pretrained)	base
Qwen3-8B	Qwen3-8B-Base	general post-training	instruct
Qwen2.5-32B	n/a	none (pretrained)	base
Qwen2.5-32B-Instruct	Qwen2.5-32B	general SFT + RLHF	instruct
OLMo-3-1025-7B (Team Olmo et al., 2025)	n/a	none (pretrained)	base
OLMo-3-7B-Instruct-SFT	OLMo-3-1025-7B	SFT	stage 1
OLMo-3-7B-Instruct-DPO	OLMo-3 SFT	DPO	stage 2
OLMo-3-7B-Instruct	OLMo-3 DPO	RLVR	stage 3 (instruct)
OLMo-3-7B-RL-Zero-General	OLMo-3-1025-7B	RL (LM judge)	RL from base
OLMo-3-1125-32B	n/a	none (pretrained)	base
OLMo-3.1-32B-Instruct-SFT	OLMo-3-1125-32B	SFT	stage 1
OLMo-3.1-32B-Instruct-DPO	OLMo-3.1 SFT	DPO	stage 2
OLMo-3.1-32B-Instruct	OLMo-3.1 DPO	RLVR	stage 3 (instruct)

Table 5: Models studied. Each pair or family shares its pretrained weights, so differences reflect post-training. OLMo 3 and OLMo 3.1 release every stage of their instruct pipelines.

Pair	Model	pass@1	pass@32	Distinct	Entropy	TV↓	Cov@32	ECE↓
Qwen3-8B, T=1	base	0.282	**0.789**	**6.60**	**1.31**	**0.712**	**0.407**	0.214
	post-trained	**0.556**	0.643	1.53	0.18	0.783	0.236	**0.061**
Qwen2.5-7B, T=1	base	0.239	**0.766**	**8.23**	**1.58**	**0.715**	**0.422**	0.303
	instruct	**0.464**	0.663	2.65	0.49	0.761	0.290	**0.137**
Qwen3-8B, T=0.7	base	0.388	**0.730**	**3.76**	**0.78**	**0.750**	**0.322**	0.121
	post-trained	**0.558**	0.607	1.27	0.11	0.793	0.217	**0.063**

Table 6: DDXPlus, 32 samples per case (980 cases). The reference differential has mean entropy 1.87. Distinct and entropy refer to the top-1 answers across samples; Cov@32 is the share of reference probability mass covered by those answers. ECE is for the stated top-1 probability.

answers but barely changes the post-trained models' (Qwen3 entropy 0.11 at T=0.7, 0.18 at T=1), so the collapse lives in the model's own distribution rather than in the decoding settings. As a result, post-trained models cover less of the reference differential: 0.24 vs. 0.41 of its probability mass at k=32 for Qwen3, and 0.29 vs. 0.42 for Qwen2.5. Their top-1 calibration, however, is markedly better (ECE 0.06 vs. 0.21 and 0.14 vs. 0.30). This concerns verbalized confidence; alignment is known to worsen the calibration of token probabilities (OpenAI, 2023; Zhu et al., 2023), while verbalized confidence of RLHF models is better calibrated (Tian et al., 2023). On every single-answer criterion (accuracy, small-k pass@k, calibration), the post-trained model therefore looks strictly better, and the loss of

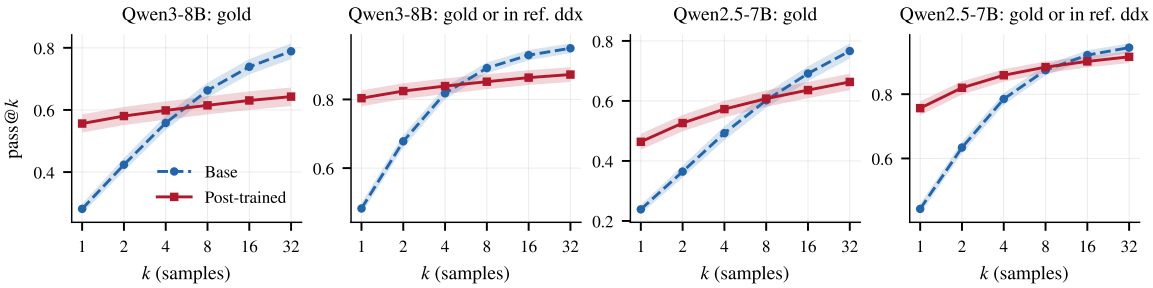


Figure 2: pass@k on DDXPlus at T=1 (980 cases, 32 samples), base model (dashed) vs. its post-trained version (solid), with 95% case-bootstrap bands. "Gold": the top-1 answer is the true pathology; "gold or in ref. ddx": it is the true pathology or appears in the reference differential. In both families the post-trained model wins at small k and loses at large k.

breadth in its samples is invisible. Whether the breadth is lost from the model's stated differential as well is a separate question (§4.2).

Which diagnoses drop out? For each case and each condition that the reference differential weights at least 0.1, we ask whether the model names that condition as its top answer in at least one of its 32 samples. We call this *considering* the condition. We group the 49 conditions by DDXPlus severity (1 = most severe). Post-training lowers consideration in every tier (Table 7), and the severe tier (DDXPlus severity 1–2, the dataset's label rather than a clinical consensus list) is not spared. This metric needs a chance reference, though: a guesser that picks uniformly among the 49 conditions scores about 0.48 on it, and one that draws from the average reference differential 0.56–0.61, above every model, base models included. Considered@32 therefore measures the breadth of the sampled answers, not clinical judgement; we assess severe conditions on the stated list instead (§E.2). For the 17 severity-1–2 conditions (e.g. NSTEMI/STEMI, acute pulmonary edema, anaphylaxis, pulmonary embolism, unstable angina), consideration falls from 0.45 to 0.27 for Qwen3 and from 0.51 to 0.34 for Qwen2.5 at T=1. The differences are -18.4 $[-20.8, -15.9]$ and -17.0 $[-19.7, -14.2]$ points. Some diagnoses disappear from the top answer almost entirely. Qwen2.5-7B-Instruct names unstable angina as its top answer in one of 107 eligible cases, where its base model does in 47%. When unstable angina is the true pathology, pass@32 falls from 1.00 to 0.00. Stable angina (0.89 to 0.06) and myocarditis (0.41 to 0.00) follow the same pattern. These conditions are not forgotten: they still appear lower in the listed differential (unstable angina in 50% of eligible cases, against 95% for the base model). The model ranks them but no longer commits to them. At the same time, post-trained models concentrate on a few salient "attractor" diagnoses. Qwen2.5-7B-Instruct gives NSTEMI/STEMI as its top answer in 13.1% of samples, against a reference rate of 4.4% and 6.4% for its base model. It also over-predicts pulmonary embolism (8.2% vs. 3.2%) and pneumonia (8.0% vs. 2.4%). Chest-pain presentations thus collapse onto the most dangerous candidate. That raises its accuracy (NSTEMI pass@1 0.99 vs. 0.44) at the cost of the alternatives. Ruling out acute coronary syndrome first is standard practice for chest pain, but these vignettes

Comparison	Considered@32 (base → post-trained) Severe (1–2)	Moderate (3)	Mild (4–5)
Qwen3-8B, T=1	0.45 → 0.27	0.50 → 0.32	0.45 → 0.20
	−18.4 [−20.8, −15.9]	−18.1 [−20.7, −15.7]	−24.2 [−27.0, −21.4]
Qwen2.5-7B, T=1	0.51 → 0.34	0.50 → 0.37	0.38 → 0.23
	−17.0 [−19.7, −14.2]	−12.1 [−14.7, −9.7]	−15.0 [−17.4, −12.6]
Qwen3-8B, T=0.7	0.34 → 0.24	0.44 → 0.29	0.31 → 0.18
	−9.8 [−11.8, −7.9]	−15.8 [−18.2, −13.5]	−13.4 [−15.8, −11.1]
Qwen2.5-7B, T=0.7	0.40 → 0.28	0.41 → 0.31	0.28 → 0.17
	−12.4 [−14.7, −10.2]	−9.1 [−11.2, −6.7]	−10.3 [−12.3, −8.3]
Instruct → AlphaMed, T=0.7	0.28 → 0.32	0.31 → 0.34	0.17 → 0.21
	+4.3 [+2.0, +6.6]	+2.2 [+0.4, +4.0]	+3.9 [+2.4, +5.4]
Instruct → HuatuoGPT-o1, T=0.7	0.28 → 0.36	0.31 → 0.37	0.17 → 0.27
	+8.4 [+6.1, +10.8]	+6.0 [+4.0, +7.9]	+9.5 [+7.5, +11.5]
Instruct → m1, T=0.7	0.28 → 0.41	0.31 → 0.41	0.17 → 0.29
	+13.5 [+10.8, +16.2]	+9.7 [+7.2, +12.2]	+11.4 [+9.2, +13.6]
Base → AlphaMed (base-RL), T=0.7	0.40 → 0.39	0.41 → 0.40	0.28 → 0.27
	−0.7 [−3.0, +1.7]	−0.4 [−2.6, +1.7]	−0.4 [−2.4, +1.7]
Base → `acc` GRPO (8B), T=0.7	0.34 → 0.31	0.44 → 0.42	0.31 → 0.27
	−2.8 [−4.4, −1.3]	−2.1 [−3.7, −0.6]	−4.3 [−6.2, −2.5]
Base → `gm` GRPO (8B), T=0.7	0.34 → 0.32	0.44 → 0.43	0.31 → 0.28
	−1.5 [−3.2, +0.1]	−1.3 [−3.0, +0.3]	−3.0 [−4.8, −1.2]

Table 7: Share of (case, condition) pairs with reference probability ≥ 0.1 in which the condition is a top-1 answer in at least one of 32 samples, by DDXPlus severity. Second rows give differences in points with 95% case-bootstrap intervals. All rows use the 980 test cases; GRPO rows are seed 0. General post-training lowers consideration in every tier, including the severity-1–2 tier; medical post-training raises it; our GRPO from the base model loses at most 4.3 points, and with the group reward the severe-tier loss is not significant.

lack the ECG and biomarkers that would separate the coronary conditions, so we read this as a concentration of the samples, not as a clinical error.

D.3. Robustness to prompt and output format

Prompt. We repeat the Qwen3-8B pair with two other prompts (Table 8). *Plain* drops the system prompt and rewords the instructions. *CoT* asks the model to reason step by step before answering; Qwen3's thinking mode stays off in all runs. The plain prompt reproduces the pattern: the post-trained model gains +23.0 [+20.9, +25.1] points at k=1 and loses −7.1 [−9.3, −5.0] at k=32 (T=0.7; at T=1, +34.2 and −8.7 [−11.0, −6.4]), with 1.3–1.5 distinct answers against 3.7–5.6. Asking for reasoning changes the picture. With CoT, the post-trained model's pass@32 rises from 0.607 to 0.835: it leads the base model at every k at T=0.7 (+4.7 [+2.9, +6.6] at k=32) and ties it at T=1 (0.0 [−2.1, +1.9]). Its samples remain far less diverse than the base model's (2.1–2.2 vs. 3.9–6.3 distinct answers), so reasoning recovers coverage at k=32 without restoring sampling diversity. The coverage loss is therefore largest when the model answers directly, and reasoning at inference time

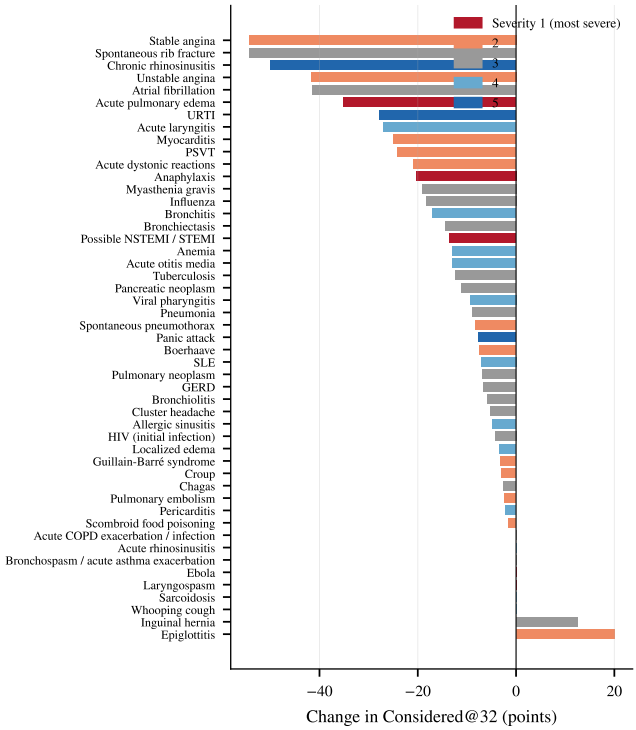


Figure 3: Change in Considered@32 per condition, Qwen2.5-7B-Instruct minus its base model (T=1, 980 cases), colored by DDXPlus severity (dark red = most severe). Conditions with fewer than five cases giving them reference probability ≥ 0.1 are omitted. Angina, acute pulmonary edema, anaphylaxis and myocarditis are among the largest drops.

mitigates it. This matches the medical models: the reasoning-trained descendants widen the differential (§H.1).

Output format. Base models more often give invalid answers: unparsable, or with a top answer outside the 49 conditions. This affects 12% of Qwen3-8B-Base samples at T=0.7 and 25% at T=1 (Qwen2.5-7B: 1% and 21%), against at most 0.3% for the post-trained models; samples that name no listed condition at all, the stricter definition used in the tables, are 11.3% and 22.3% (Qwen2.5-7B: 0.9% and 18.4%). Invalid samples count as wrong, which deflates the base model at large k but inflates the post-trained model's gain at small k. Among valid samples only, the post-trained models still gain +12.0 $[+9.8, +14.1]$ (Qwen3, T=0.7), +18.7 $[+16.6, +20.7]$ (Qwen3, T=1), +12.2 $[+10.4, +14.0]$ and +16.8 $[+15.0, +18.5]$ points (Qwen2.5) at k=1, so a quarter to a third of the raw gain at T=1 is format. On the cases where every sample of both models is valid, the large-k loss persists: −15.2 $[-21.9, -8.6]$ at k=32 for Qwen3 (105 cases) and −10.2 $[-12.7, -7.6]$ for Qwen2.5 (748 cases), both at T=0.7. (At T=1 almost no case has 32 valid base samples.) The reported large-k losses are thus lower bounds.

		pass@1		pass@32		Distinct	
Prompt	T	Base	Post	Base	Post	Base	Post
default	0.7	0.388	**0.558**	**0.730**	0.607	3.76	1.27
default	1.0	0.282	**0.556**	**0.789**	0.643	6.60	1.53
plain	0.7	0.352	**0.583**	**0.727**	0.655	3.72	1.28
plain	1.0	0.244	**0.586**	**0.792**	0.705	5.57	1.54
CoT	0.7	0.409	**0.652**	0.788	**0.835**	3.89	2.08
CoT	1.0	0.275	**0.642**	0.837	0.837	6.31	2.24

Table 8: Qwen3-8B-Base vs. Qwen3-8B on DDXPlus (980 cases, 32 samples) under three prompts. Bold marks the significantly better model (paired case bootstrap). Without reasoning, the base model wins at k=32; with CoT, the post-trained model matches or beats it, but its samples stay far less diverse.

D.4. Which post-training stage narrows the differential?

The Qwen pairs compare a base model with the end of a post-training pipeline. OLMo 3 (Team Olmo et al., 2025) releases every stage of its 7B instruct pipeline from one base model: supervised fine-tuning, then DPO, then RL with verifiable rewards on math, code, instruction-following and chat prompts (the released OLMo-3-7B-Instruct). It also releases RL-Zero checkpoints trained with RL directly from the base; we use the general one, trained on chat prompts with an LM-judge reward. (The RL-Zero-Mix checkpoint does not load in our vLLM version.) Table 9 gives the results.

The coverage loss replicates in OLMo. At T=0.7, every post-trained stage is worse than the base at k=32: by -14.6 $[-17.2, -11.6]$ points after SFT, -9.2 $[-12.1, -6.1]$ after DPO, -28.3 $[-31.4, -25.1]$ after RLVR, and -13.3 $[-16.1, -10.5]$ for RL-Zero. Diversity falls at every stage of the pipeline, from 4.93 distinct answers for the base to 4.01 after SFT, 3.30 after DPO and 2.37 after RLVR.

Compared stage to stage, DPO recovers part of what SFT lost at large k ($+5.4$ $[+2.9, +8.1]$ points at k=32, and $+10.3$ $[+8.9, +11.6]$ at k=1) while narrowing the differential further (-0.71 distinct answers). The final RLVR stage is the largest single narrowing. Against the DPO model it loses -19.1 $[-21.9, -16.5]$ points at k=32, 0.92 distinct answers and 7.7 points of coverage@32, and it is also worse at k=1 (-5.9 $[-7.1, -4.7]$). It drops severe-tier Considered@32 to 0.25, a loss of -16.1 $[-18.6, -13.5]$ points relative to the base, against -7.4 for the DPO model. At T=1 the pattern is the same (RLVR vs. DPO: -12.7 $[-15.4, -9.9]$ at k=32).

RL applied directly to the base narrows the differential less. At T=1, RL-Zero shows no detectable loss of the base model's pass@32 (-1.8 $[-4.4, +0.7]$; clustered $[-5.8, +2.1]$) and 6.28 of its 7.53 distinct answers, and it loses 4.0 $[1.2, 6.9]$ points of severe-tier consideration, against 14.4 for the full instruct pipeline. At T=0.7 it still loses 13.3 points at k=32 (clustered $[7.3, 19.7]$), about as much as SFT.

The OLMo base model often breaks the output format: 40% of its samples are invalid at T=0.7 and 47% at T=1, while unparsable completions are 16–21% for the SFT model and at most 1.6% for the later stages. No case has 32 valid base samples, so the all-valid

T	Stage	pass@1	pass@32	Distinct	Entropy	Cov@32	Severe C@32
0.7	Base	0.178	**0.705**	**4.93**	**1.16**	**0.351**	**0.412**
	SFT	0.222	0.559	4.01	0.87	0.257	0.373
	SFT+DPO	**0.324**	0.613	3.30	0.63	0.272	0.338
	SFT+DPO+RLVR (Instruct)	0.265	0.422	2.37	0.42	0.195	0.251
	RL-Zero (RL from base)	0.295	0.572	4.06	0.84	0.286	0.326
1.0	Base	0.128	**0.713**	**7.53**	**1.66**	**0.398**	**0.469**
	SFT	0.191	0.634	6.32	1.29	0.321	0.426
	SFT+DPO	**0.313**	0.688	4.48	0.87	0.312	0.410
	SFT+DPO+RLVR (Instruct)	0.272	0.561	3.58	0.68	0.259	0.325
	RL-Zero (RL from base)	0.279	0.695	6.28	1.21	0.367	0.429

Table 9: OLMo 3 7B post-training stages on DDXPlus (980 cases, 32 samples). Severe C@32 is severe-tier Considered@32 as in Table 7. Bold marks the best value in each block. Diversity falls at every stage of the instruct pipeline, and the RLVR stage loses the most coverage; RL from the base keeps the base model's pass@32 at T=1.

check of §D.3 is not possible here. Among valid samples, the post-trained models' gains at k=1 are small (-1.9 to $+4.0$ points at T=0.7, $+1.3$ to $+8.1$ at T=1), so most of OLMo's raw gain at k=1 is format. Invalid samples count as misses and are excluded from distinct answers and entropy, so the base model's coverage and diversity are understated and the losses above are conservative.

Across these pipelines, then, the collapse of the sampled differential is not a necessary effect of RL. It accumulates over the instruction-tuning pipeline; in OLMo 3 7B its largest step is RL applied on top of SFT and DPO, but at 32B it is SFT (§D.5), so which stage narrows most depends on the recipe. RL applied directly to a base model (OLMo RL-Zero at T=1, AlphaMed's base-initialized checkpoint, and our GRPO runs in §I) narrows the differential less, but not always: RL-Zero at T=0.7 loses about as much as SFT. On Phenopacket Store the instruct pipeline also loses coverage: every stage trails the base model at k=32 (-2.0 [$-3.0, -1.1$] points after SFT, -1.5 after DPO, -1.4 after RLVR; all significant), while RL-Zero does not (-0.6 [$-1.4, +0.2$]), from a base model at 4.5% (§H.2).

D.5. Replication at 32B

We repeat the DDXPlus comparisons at T=0.7 with Qwen2.5-32B and its Instruct model, and with every released stage of OLMo 3.1 32B, whose base is OLMo-3-1125-32B; there is no 32B RL-Zero checkpoint (Table 10). The sampled differential collapses at 32B too, by about half as much. Post-training costs 4.9 [2.9, 7.1] points at pass@32 for Qwen2.5-32B and 4.0 [2.0, 6.0] for OLMo 3.1 32B (clustered [-0.5, 10.3] and [1.2, 6.8]; neither survives correction), halves the number of distinct answers (3.78 $\rightarrow$ 1.68 and 4.32 $\rightarrow$ 2.15), and lowers severe-tier consideration by 6.1 and 8.4 points. For Qwen the loss holds on the 421 cases where every base sample is valid (-4.8 [$-7.8, -1.7$]); the OLMo 32B base gives invalid answers on 27% of samples. With few-shot prompts, which nearly remove invalid answers, both losses are three to five times larger (§F.1). The stage attribution differs from 7B. In OLMo 3.1 32B

Model	pass@1	pass@32	Distinct	Cov@32	Severe C@32	TV one list↓	TV samples↓	Vote@32 gain
Qwen2.5-32B (base)	0.500	**0.795**	**3.78**	**0.352**	**0.35**	0.726	**0.726**	+7.0
Qwen2.5-32B-Instruct	**0.604**	0.746	1.68	0.276	0.29	**0.674**	0.759	+1.9
OLMo-3-1125-32B (base)	0.326	**0.796**	**4.32**	**0.347**	**0.39**	0.746	**0.742**	+8.4
SFT	0.461	0.674	2.94	0.265	0.27	0.736	0.776	+3.6
SFT+DPO	**0.584**	0.784	2.29	0.293	0.32	**0.694**	0.755	+2.1
SFT+DPO+RLVR (Instruct)	0.560	0.756	2.15	0.283	0.30	0.701	0.763	+0.9

Table 10: 32B replication on DDXPlus (T=0.7, 980 cases, 32 samples). TV columns: distance to the reference differential of one stated list and of the sampled top-1 answers (§4.2). Vote@32 gain: accuracy of the majority answer over 32 samples minus single-call accuracy, in points (§4.3). Bold: best value per family.

the largest narrowing is SFT (pass@32 −12.1 [−14.5, −9.8], clustered [−18.0, −6.8], distinct answers −1.39, severe tier −11.9); DPO recovers most of it (+10.9 against SFT), and the final RL step is small (−2.8 [−4.4, −1.2] at k=32, clustered [−5.3, −0.5], −0.14 distinct answers). OLMo 3.1 32B and OLMo 3 7B were post-trained with different recipes, so the stage that narrows the samples most is a property of the pipeline, not of SFT, DPO or RL as such. Under this prompt the stated readout improves in both 32B families (§4.2; with few-shot prompts only Qwen2.5-32B's does robustly, §F.1), and voting stops paying off: majority voting over 32 samples gains the base models 7.0 and 8.4 points and the Instruct models 1.9 and 0.9 (the latter n.s.).

Appendix E. The stated readout: elicitation and robustness

E.1. Robustness of the stated readout to elicitation

The stated readout above uses a five-item list. We repeat the Qwen pairs and OLMo 3 7B's base and Instruct models at T=0.7 with a prompt that asks for a probability on every one of the 49 conditions, summing to one, and take each sample's top answer to be its highest-probability condition (Table 11). The results match the short list for Qwen. After post-training the stated distribution is closer to the reference (TV −0.052 for Qwen3-8B, −0.067 for Qwen2.5-7B-Instruct, −0.032 for OLMo-3-7B-Instruct) while the samples move away (+0.012 to +0.038); OLMo-3-7B-Instruct's full distribution moves closer although its five-item list does not. Qwen3-8B's stated probability again beats agreement over 32 samples (+0.069 [+0.033, +0.104]), Qwen2.5-7B-Instruct's again does not (−0.074), and OLMo-3-7B-Instruct's ties it (−0.010; +0.044 against five samples). The top three of one full distribution beats the three most frequent of 32 sampled answers for Qwen3-8B (+14.6 points) and narrowly for Qwen2.5-7B-Instruct (+1.6), but not for OLMo-3-7B-Instruct (−9.5).

E.2. Ruling out format, accuracy and floor effects

Table 13 tests simpler explanations of the stated-list result. For reference, one list's recall@5 rises after post-training by 13.6, 16.2 and 12.0 points for Qwen3-8B, Qwen2.5-7B and Qwen2.5-32B at T=0.7 and by 21.9 and 24.0 for the 7–8B Qwen models at T=1, by

Model	TV to reference↓ One distribution	TV to reference↓ Samples	AUROC, single-call answer correct Stated p	AUROC, single-call answer correct Agree@5	AUROC, single-call answer correct Agree@32	Top-3: one dist. − 32 samples
Qwen3-8B-Base	0.733	**0.752**	0.647	0.809	**0.845**	−6.7
Qwen3-8B	**0.681**	0.790	**0.717**	0.609	0.647	+14.6
Qwen2.5-7B	0.769	**0.758**	0.688	0.792	**0.827**	−10.2
Qwen2.5-7B-Instruct	**0.702**	0.770	0.674	0.715	**0.748**	+1.6
OLMo-3-1025-7B	0.807	**0.766**	0.610	0.744	**0.786**	−15.3
OLMo-3-7B-Instruct	**0.775**	0.787	0.749	0.705	**0.759**	−9.5

Table 11: Full-distribution prompt on DDXPlus (T=0.7, 980 cases): a probability on every condition, top answer = highest stated probability. TV: one sample's stated distribution vs. the sampled top-1 answers (bold: better model of each pair). AUROC: bold marks each model's best signal.

12.7 for OLMo 3.1 32B, and for OLMo 3 7B by 1.5 (n.s.) at T=0.7 and 10.0 at T=1 (condition-clustered intervals exclude zero except where marked). *Format.* The general models' post-trained lists and base-model lists at T=0.7 are nearly all complete (81–99% name five on-list conditions with probabilities; Qwen base models at T=1, 61–62%; the medical models, 61–80%). Restricting to complete lists leaves the gaps unchanged (Qwen2.5-7B: 0.779 vs. 0.719). *Accuracy.* A more accurate model puts more stated mass on the true pathology, which the reference also favors. Within samples whose top answer is correct, and within those whose top answer is wrong, the post-trained Qwen list is still closer to the reference (−0.018 to −0.034 and −0.036 to −0.079 in TV across both temperatures; Table 13 shows T=0.7). *Floor.* One list can do no better than the reference's own top five, and a finite set of samples no better than as many draws from the reference. Measured as excess over these floors, post-training lowers the list's TV (Qwen3-8B 0.479 → 0.435) and raises the samples' (0.557 → 0.613). *True pathology removed.* With the true pathology removed from both list and reference and both renormalized, the post-trained Qwen and OLMo 3.1 32B lists are still closer to the reference (−0.018 to −0.056), and DDXPlus's set F1 against the reference's support improves by 0.040–0.083 (Table 34). TV, mass and DDF1 are not independent, since a list's TV is the reference mass it leaves out plus the mass it misallocates among the conditions it names, and the second term grows mechanically with the first. The conditional TV, between the list and the reference restricted to the named conditions and renormalized, removes this: it falls after post-training for every Qwen pair and OLMo 3.1 32B (−0.025 to −0.078, all surviving correction) and rises for OLMo 3 7B at T=1 (+0.064), so the Qwen lists both name better conditions and weight them more like the reference. *Patient-blind list.* A fixed list of the five conditions with the highest average reference probability scores TV 0.794, mass 0.238 and DDF1 0.272. Every model's list beats it in TV except OLMo 3 7B's and Qwen2.5-7B's base list at T=1; in DDF1 several base lists do not (Qwen2.5-7B 0.245, OLMo 3 7B 0.225), and the post-trained Qwen and OLMo 3.1 32B lists exceed it by 0.035–0.061. *Over-naming.* As a share of named conditions, post-trained lists name fewer that the reference gives less than 1% (Qwen3-8B 0.59 → 0.55) and fewer unwarranted severe conditions (Qwen2.5-32B 0.74 → 0.65), but as a count per list the change is mixed: from 0.22 fewer unwarranted severe conditions

Pair (base → post)	Named less / more often	Significant less / more	P(as many less often)	Mean shift (17 conditions)	P	Largest loss (shift)
Qwen3-8B, T=0.7	5 / 12	2 / 3	0.98	+0.040	0.001	acute pulmonary edema (-0.14)
Qwen3-8B, T=1	6 / 11	2 / 4	0.94	+0.060	0.001	stable angina (-0.06)
Qwen2.5-7B, T=0.7	5 / 12	2 / 2	0.98	+0.029	0.001	stable angina (-0.16)
Qwen2.5-7B, T=1	4 / 13	2 / 5	0.99	+0.065	0.001	stable angina (-0.13)
Qwen2.5-32B, T=0.7	6 / 11	3 / 5	0.93	+0.047	0.001	acute pulmonary edema (-0.19)
OLMo 3 7B, T=0.7	8 / 9	7 / 6	0.68	+0.046	0.001	acute pulmonary edema (-0.24)
OLMo 3 7B, T=1	8 / 9	6 / 6	0.68	+0.061	0.001	NSTEMI/STEMI (-0.12)
OLMo 3.1 32B, T=0.7	7 / 10	5 / 0	0.82	+0.001	0.876	acute pulmonary edema (-0.16)
Qwen3-8B, few-shot	5 / 12	1 / 4	0.97	+0.061	0.001	acute pulmonary edema (-0.16)
Qwen2.5-7B, few-shot	8 / 9	4 / 2	0.69	+0.034	0.001	stable angina (-0.15)
Qwen2.5-32B, few-shot	10 / 7	5 / 3	0.29	−0.012	0.009	laryngospasm (-0.25)
OLMo 3 7B, few-shot	9 / 8	7 / 3	0.51	−0.018	0.018	croup (-0.51)
OLMo 3.1 32B, few-shot	9 / 8	4 / 2	0.50	+0.011	0.012	stable angina (-0.25)
Mistral-7B, few-shot	11 / 6	3 / 0	0.17	−0.009	0.052	pulmonary embolism (-0.31)

Table 12: The 17 severe conditions one by one (DDXPlus). For each condition, the share of a model's lists that name it, over the cases whose reference gives it any weight; post-trained minus base, with intervals that resample the true pathologies (50,000 resamples for the zero-shot rows, 10,000 for the few-shot rows) and Benjamini–Hochberg correction across the 17 conditions. P columns compare the count of conditions named less often and the mean shift across conditions with a null that swaps each case's base and post-trained lists at random (2,000 swaps); the count test is weak against an overall upward shift, so the mean shift and the largest single loss are shown too.

(OLMo 3.1 32B) to 0.35 more (OLMo 3 7B). *Severe conditions one by one.* Table 12 tests each of the 17 severe conditions, counting it wherever the reference gives it any weight. After Benjamini–Hochberg correction, 2–7 conditions per zero-shot pair are named significantly less often after post-training and 0–6 significantly more often, and no pair names more conditions less often than a random swap of base and post-trained lists would. With the warranted threshold at 0.01 or 0.05 instead of 0.1, post-trained lists still name more severe conditions in seven of eight zero-shot settings (+2.5 to +8.1 points; OLMo 3.1 32B n.s.). That count test is weak against an overall upward shift, so Table 12 also gives the mean shift across the 17 conditions: it rises significantly in seven of eight zero-shot settings but falls with worked examples for Qwen2.5-32B (−0.012) and OLMo 3 7B (−0.018), and is flat for Mistral (−0.009, p=0.05). Significant losses include acute pulmonary edema (Qwen3-8B 0.75 → 0.61, Qwen2.5-32B 0.63 → 0.44, OLMo 3 7B 0.56 → 0.31, OLMo 3.1 32B 0.57 → 0.41), stable angina (Qwen2.5-7B 0.25 → 0.08), NSTEMI/STEMI (OLMo 3 7B 0.31 → 0.14, OLMo 3.1 32B 0.49 → 0.36) and myocarditis (Qwen2.5-32B 0.14 → 0.05); significant gains include pulmonary embolism (Qwen2.5-7B 0.29 → 0.56, OLMo 3 7B 0.20 → 0.65), NSTEMI/STEMI (Qwen2.5-7B 0.32 → 0.52) and unstable angina (Qwen3-8B 0.14 → 0.37). Among single stages, OLMo 3 7B's RLVR step and OLMo 3.1 32B's SFT and RLVR steps do name more conditions less often than chance (12–13 of 17, p=0.03–0.04).

Sampling from the list. Replacing each sample's top answer with a draw from its own stated list raises pass@32 for every model. Compared like for like, post-trained lists reach higher than their base model's lists in five of seven comparisons (Qwen3-8B 0.811 vs. 0.780

Model (T=0.7)	Complete lists	TV of one list↓ complete only	top-1 right	top-1 wrong	Excess TV over floor↓ One list	Samples	Severe named ref. < 0.1	pass@32 list-sampled
Qwen3-8B-Base	0.90	0.743	0.663	0.806	0.479	**0.557**	0.752	0.780
Qwen3-8B	0.99	**0.701**	**0.646**	**0.769**	**0.435**	0.613	**0.680**	**0.811**
Qwen2.5-7B	0.95	0.779	0.689	0.828	0.514	**0.575**	0.823	0.772
Qwen2.5-7B-Instruct	0.91	**0.719**	**0.655**	**0.773**	**0.452**	0.605	**0.796**	**0.802**
Qwen2.5-32B	0.96	0.725	0.661	0.795	0.462	**0.543**	0.735	0.850
Qwen2.5-32B-Instruct	0.98	**0.674**	**0.639**	**0.726**	**0.409**	0.579	**0.645**	**0.880**
OLMo-3-1025-7B	0.81	**0.794**	0.687	**0.837**	**0.524**	**0.522**	**0.747**	**0.719**
OLMo-3-7B-Instruct	0.95	0.823	**0.659**	0.882	0.558	0.671	0.779	0.577
OLMo-3-1125-32B	0.89	0.744	0.662	0.812	0.481	**0.532**	0.706	0.839
OLMo-3.1-32B-Instruct	0.99	**0.701**	**0.644**	**0.774**	**0.436**	0.583	**0.634**	**0.853**

Table 13: Robustness of the stated-list result (DDXPlus, T=0.7, 980 cases, valid samples). Complete lists: share naming five on-list conditions with valid probabilities. TV of one list on complete lists only, and on samples whose top answer is right or wrong. Excess TV: TV minus the expected TV of the matching oracle (the reference's top five for one list; as many reference draws as the model has valid samples for the samples). Severe named, ref. < 0.1: share of named severe conditions that the reference gives less than 0.1 (over-naming). pass@32 list-sampled: top answers drawn from each sample's stated list. Bold: better model of each pair. Excess TV is averaged over samples against each case's floor, so it is not exactly the difference of the mean TV and the mean floor. A uniform guesser scores about 0.48 at pass@32.

at T=0.7, Qwen2.5-32B-Instruct 0.880 vs. 0.850, OLMo 3.1 32B Instruct 0.853 vs. 0.839), but not for Qwen3-8B at T=1 (0.820 vs. 0.832) or OLMo 3 7B (0.577 vs. 0.719), and a uniform guesser reaches 0.48. OLMo 3 7B remains the exception throughout: its Instruct list is closer to the reference when its top answer is correct but much farther when it is wrong.

Appendix F. Controls: prompt, temperature, decoding, severity labels and a third family

This section collects the controls for the main comparison. Table 14 shows, for each control, whether each main claim holds; the subsections give the numbers. Real-case evidence is in Appendix H.

F.1. Few-shot elicitation

The main analyses give every model the same zero-shot chat prompt, which base models were not trained to follow; they give invalid answers on up to 47% of samples, which we count as wrong. We therefore re-ran the five final base/post-trained pairs at T=0.7 with three worked examples in the prompt: three fixed DDXPlus training patients with different true pathologies, each followed by the top five conditions of its reference differential, with probabilities, in the JSON format the instructions ask for. Base models see the instructions and examples as a plain document and complete the next one (generation stops at the end of

Control	(a) View kept, variation removed	(b) Samples collapse	(c) First-answer gain is sampling noise	(d) Stated list improves (smaller Qwen)	(e, f) Severe conditions; must-not-miss prompt	(g) Voting and agreement weaken
Worked examples, reference lists (§F.1)	holds	holds, larger (15–30 pts)	–	holds for Qwen3-8B, Qwen2.5-7B; TV only for Qwen2.5-32B; weaker for OLMo 3.1 32B	partly: mean falls for Qwen2.5-32B, OLMo 3 7B	holds
Worked examples from another model (§F.2)	–	holds (13–30 pts)	–	holds for Qwen and Mistral, not OLMo	no excess losses (count test)	holds
Matched temperature (§F.2)	–	Qwen3-8B holds; Qwen2.5-7B-Instruct matches at T=1.3	–	holds	–	same split as (b)
Greedy decoding, matched prompts (§F.3)	–	–	holds in 4 of 6 pairs; not Qwen3-8B, nor Mistral with shared examples	holds for Qwen3-8B, Qwen2.5-7B (not Qwen2.5-32B recall); OLMo 3 7B farther	–	–
Temperature frontier (§F.3)	–	holds for Qwen3-8B, OLMo 3 7B; curves meet for Qwen2.5-7B	–	list unchanged across T	–	agreement below base at every T
Must-not-miss prompt (§F.3)	–	–	–	–	coverage +6–15 pts, partly by length	–
Published must-not-miss sets (§F.4)	–	–	–	–	mean up in 7 of 8; specific losses only OLMo, Mistral, few-shot Qwen2.5	–
Format, accuracy, floor, complete lists (§E.2)	–	–	–	holds	–	–
Other families' base models (Table 27)	holds across families; blurs within	–	–	–	–	–
Every other model's view (Table 3)	partly: own base closest for 3 of 5; as close to nearest post-trained model	–	–	–	–	–
Chance-corrected pass@32 (§F.5)	–	holds with examples (7–18 pts); zero-shot 32B loss vanishes	–	–	–	–
Ties, list size, validity, prevalence (§F.6)	–	holds at natural prevalence	–	holds at natural prevalence	–	agreement weaker without ties; top-3 list advantage is list size
Concentration, misses, coverage, common cases (§F.7)	–	holds on common cases	–	lists do not concentrate (Qwen, OLMo 3.1 32B)	fewer per-case misses	holds on common cases and at 50% coverage
32B pairs (§D.5)	holds	n.s. zero-shot; holds with examples	holds (both pairs)	holds for Qwen; partly OLMo	holds for Qwen; OLMo n.s.	partly: agreement drop n.s. for OLMo
Mistral-7B, third family (§F.8)	holds	holds	fails with shared examples	holds, smallest gain	no mean gain; 3 of 17 lost significantly	holds
Thinking mode (§G.1)	holds	fails: pass@32 restored, still narrow	–	holds, closest list	–	fails: agreement better again
Rare-disease cases (§H.4)	–	–	–	–	–	holds; agreement stays the better signal
Published case reports (§H.3)	–	holds for the authors' alternatives (Qwen); pass@32 for 1 of 4	–	holds (recall +2 to +6; alternatives unchanged)	–	partly: voting gains small for all; stated p catches up with agreement

Table 14: Which claims each control supports. "Holds": the control reproduces the claim; "partly": in some pairs or settings; "fails": it reverses or disappears; –: not tested. Claims are lettered as in Table 4, which lists the evidence claim by claim; (h) is tested only in §G.3.

the list); post-trained models see the same instructions and examples as earlier chat turns. Invalid samples fall below 0.5% for every model (Table 15).

The examples help base models most. Their pass@1 rises from 0.178 to 0.306 for OLMo-3-1025-7B, from 0.326 to 0.528 for OLMo-3-1125-32B and from 0.500 to 0.647 for Qwen2.5-32B, and their pass@32 reaches 0.76–0.90, while their stated lists barely change (TV 0.712–0.787, against 0.726–0.794 zero-shot). The losses in the samples therefore grow (Table 16): post-training costs 15.2–29.9 points of pass@32 in every pair, including both 32B pairs, whose zero-shot losses did not survive correction, and every post-trained model's Cov@32 (0.21–0.25) is below every base model's (0.33–0.45) and below a uniform guesser's.

For the three Qwen pairs the stated-list improvement persists and mostly grows: TV -0.057 to -0.087, -0.055 to -0.073 with the true pathology removed, and DDF1 $+0.051$ to $+0.081$, all surviving correction. OLMo 3 7B remains the exception (TV $+0.021$, n.s.; $+0.024$ without the true pathology). For OLMo 3.1 32B the improvement shrinks: the post-trained list still places more reference mass on the conditions it names ($+0.047$), but its TV (-0.022), TV without the true pathology (-0.006) and DDF1 ($+0.009$) no longer survive correction. At 32B the stated-list result is therefore robust for Qwen2.5-32B but depends on the prompt for OLMo 3.1 32B. Post-trained lists still name 5–10 of the 17 severe conditions less often than their base model's.

The consequences replicate in all five pairs. Majority voting over 32 samples gains base models 6.2–10.9 points and post-trained models 0.2–1.1. The three most frequent of 32 sampled answers beat the top three of one stated list by 8.2–22.0 points for base models, while for post-trained models the list is ahead by 0.2–16.7. Agreement with four more samples is the better error signal for every base model (AUROC 0.79–0.84, against 0.67–0.73 for the stated probability), and the stated probability for every post-trained model (0.69–0.77, against 0.56–0.68); against agreement across 31 samples the stated probability wins clearly only for Qwen3-8B and Qwen2.5-7B-Instruct ($+0.108$ and $+0.130$, clustered) and ties for the others. This check uses one temperature and one set of examples, and the examples show the reference's own lists, which may help every model match the reference.

F.2. Two further controls

Reference-free few-shot examples. The examples of §F.1 show each training patient's reference top five, and so also the target of the evaluation. We re-ran the six final pairs with the same three patients but with each example list replaced by Qwen2.5-32B-Instruct's own greedy answer (on-list, with the true pathology first for two patients and second for one). The conclusions do not change (Table 17). Post-training costs 12.7–29.7 points of pass@32 in every pair. The stated list moves closer to the reference for the three Qwen pairs and, now clearly, for Mistral (TV -0.053 to -0.077; -0.031 to -0.059 with the true pathology removed), but not for OLMo 3 7B or OLMo 3.1 32B, and wherever the list improves its allocation among the named conditions also moves closer to the reference (conditional TV -0.051 to -0.100). Voting helps base models by 3.7–13.7 points and post-trained ones by at most 1.4; the top three of one list beats the three most frequent samples for every post-trained model; the stated probability beats agreement as an error signal for every post-trained model and agreement beats it for every base model; and no pair names more

Model	Invalid (%)	pass@1	pass@32	Cov@32	TV, one list all	TV, one list w/o true	DDF1	TV samples	Vote@32 gain	Top-3 list − samples	AUROC stated	AUROC agree@5	AUROC agree@32
Qwen3-8B-Base	0.1	0.403	**0.809**	**0.383**	0.768	0.860	0.245	**0.732**	+8.4	−16.0	0.722	0.828	0.862
Qwen3-8B	0.0	**0.601**	0.639	0.221	**0.682**	**0.805**	**0.327**	0.786	+0.2	+14.5	0.693	0.563	0.584
Qwen2.5-7B (base)	0.4	0.398	**0.761**	**0.333**	0.784	0.879	0.238	**0.756**	+6.2	−15.6	0.734	0.792	0.827
Qwen2.5-7B-Instruct	0.0	**0.450**	0.535	0.220	**0.702**	**0.806**	**0.309**	0.798	+0.6	+16.7	0.757	0.594	0.626
Qwen2.5-32B (base)	0.1	0.647	**0.896**	**0.343**	0.712	0.847	0.298	**0.715**	+7.1	−8.2	0.726	0.838	0.875
Qwen2.5-32B-Instruct	0.0	**0.663**	0.744	0.252	**0.654**	**0.775**	**0.349**	0.758	+0.6	+10.1	0.718	0.647	0.694
OLMo-3-1025-7B (base)	0.2	0.306	**0.857**	**0.454**	**0.787**	**0.859**	**0.245**	**0.712**	+10.9	−22.0	0.674	0.823	0.871
OLMo-3-7B-Instruct	0.1	**0.338**	0.558	0.213	0.808	0.884	0.238	0.830	+0.7	+0.2	0.733	0.668	0.696
OLMo-3-1125-32B (base)	0.4	0.528	**0.888**	**0.397**	0.719	0.818	0.302	**0.699**	+8.9	−13.4	0.670	0.826	0.865
OLMo-3.1-32B-Instruct	0.0	**0.575**	0.697	0.249	**0.697**	**0.812**	**0.311**	0.772	+1.1	+3.7	0.772	0.684	0.744

Table 15: Few-shot elicitation on DDXPlus (T=0.7, 980 cases, 32 samples; three worked examples in the prompt for every model, §F.1). Invalid: samples naming no listed condition. TV: distance to the reference differential, for one stated list (with the true pathology removed from both in "w/o true") and for the sampled top answers. Vote@32 gain and top-3 list minus top-3 of the samples are in accuracy points. AUROC: for the single-call answer being correct, of its stated probability and of agreement with the next 4 or the next 31 samples. Bold: better of the pair.

Pair (base → post)	pass@32 (points)	TV samples	TV, one list all	TV, one list w/o true	DDF1	Reference mass of one list
Qwen3-8B	−12.4 / −17.0	+0.043 / +0.054	−0.043 / −0.087	−0.018† / −0.055	+0.042 / +0.081	+0.062 / +0.113
Qwen2.5-7B	−9.7 / −22.7	+0.028† / +0.042	−0.062 / −0.082	−0.041 / −0.073	+0.062 / +0.072	+0.073 / +0.124
Qwen2.5-32B	−4.9† / −15.2	+0.033 / +0.044	−0.053 / −0.057	−0.043 / −0.072	+0.051 / +0.051	+0.067 / +0.084
OLMo 3 7B	−28.3 / −29.9	+0.090 / +0.118	+0.029† / +0.021†	+0.034 / +0.024	−0.006† / −0.007†	−0.010† / −0.002†
OLMo 3.1 32B	−4.0† / −19.1	+0.020† / +0.074	−0.045 / −0.022†	−0.023 / −0.006†	+0.040 / +0.009†	+0.056 / +0.047

Table 16: Base → post-trained differences under the zero-shot prompt / the few-shot prompt (DDXPlus, T=0.7). †: not significant under the condition-clustered bootstrap with Holm correction (over 112 tests for the zero-shot and 70 for the few-shot pairs; Appendix J).

severe conditions less often than chance. Five of six base models score lower at k=1 than with reference-derived examples (OLMo-3-1125-32B 0.453 against 0.528).

Matched temperature. Post-trained models are sharper than their base models at the same nominal temperature, so part of the narrowing could be colder effective sampling. We sampled Qwen3-8B and Qwen2.5-7B-Instruct at T=1.3 and 1.6 (Table 18), and the two models differ. Qwen3-8B barely broadens: at T=1.6 it gives 1.92 distinct answers and reaches 0.684 at pass@32, still below its base model at T=0.7 (0.732) and at T=1 (0.789; −10.5 points, surviving correction); voting gains stay under 2 points, agreement stays weaker than for the base model, and its stated list does not change (TV 0.701). Qwen2.5-7B-Instruct does broaden: at T=1.3, with 2% invalid answers, it gives 3.38 distinct answers, matches its base model's pass@32 at T=0.7 (0.704 against 0.695), gains 5.6 points from voting, and agreement is again the better error signal (0.794 against 0.602), at a cost of 4.2 points of pass@1; at T=1.6, 31% of its answers are invalid. Its stated list stays closer to the reference than the base model's (0.722 against 0.779). For some models, then, much of the

Pair (base → post)	pass@32 (points)	TV samples	TV, one list all	TV, one list w/o true	DDF1	Cond. TV
Qwen3-8B	−14.6	+0.049	−0.077	−0.049	+0.079	−0.078
Qwen2.5-7B	−19.7	+0.031†	−0.077	−0.059	+0.074	−0.094
Qwen2.5-32B	−14.5	+0.036	−0.053	−0.057	+0.054	−0.051
OLMo 3 7B	−29.7	+0.116	−0.007†	+0.013†	−0.002†	−0.015†
OLMo 3.1 32B	−17.4	+0.073	−0.016†	+0.002†	+0.009†	−0.005†
Mistral-7B	−12.7	+0.038†	−0.059	−0.031	+0.057	−0.100

Table 17: Few-shot control with reference-free examples (DDXPlus, T=0.7, 980 cases): the same three training patients as in §F.1, but each example list is Qwen2.5-32B-Instruct's own greedy answer rather than the reference's top five. Post-trained minus base. †: not significant (condition-clustered bootstrap, Holm over 84 tests; conditional TV Holm-corrected as a separate family).

Model	T	Invalid (%)	pass@1	pass@32	Distinct	TV samples	TV one list	Vote@32 gain	AUROC stated	AUROC agree@32
Qwen3-8B-Base	0.7	11.3	0.388	0.732	3.77	0.750	0.743	+3.4	0.595	0.810
	1	22.3	0.282	0.789	6.60	0.715	0.775	+10.4	0.597	0.857
Qwen3-8B	0.7	0.0	0.559	0.607	1.27	0.793	0.700	+0.3	0.734	0.655
	1	0.0	0.556	0.643	1.53	0.783	0.700	+1.4	0.726	0.701
	1.3	0.0	0.552	0.668	1.73	0.776	0.701	+1.0	0.707	0.724
	1.6	0.0	0.549	0.684	1.92	0.770	0.701	+1.8	0.715	0.745
Qwen2.5-7B	0.7	0.9	0.352	0.695	4.91	0.757	0.779	+5.2	0.576	0.811
	1	18.4	0.239	0.766	8.23	0.719	0.805	+14.1	0.587	0.867
Qwen2.5-7B-Instruct	0.7	0.0	0.478	0.598	1.82	0.785	0.717	+0.8	0.571	0.702
	1	0.0	0.464	0.663	2.65	0.761	0.717	+4.4	0.583	0.767
	1.3	2.2	0.436	0.704	3.38	0.745	0.722	+5.6	0.602	0.794
	1.6	30.5	0.304	0.726	3.88	0.729	0.749	+10.0	0.637	0.814

Table 18: Matched-temperature control (DDXPlus, 980 cases, 32 samples, zero-shot prompt): the post-trained models sampled at higher temperatures (top-p 1) against their base models at T=0.7 and T=1. Columns as in Tables 1, 2 and 29.

sampled narrowing is sharper sampling that a higher temperature undoes; the stated-list result is not.

F.3. Greedy decoding, temperature frontier and a must-not-miss prompt

Greedy single call. Deployed systems often decode greedily. We compared one greedy answer per case under three prompt pairings (Table 19): both arms with the reference-derived worked examples of §F.1 (*same examples*); base models with the reference-free examples of Table 17 against zero-shot post-trained models, so that neither arm sees a reference list (*reference-free base*); and reference-example base models against zero-shot post-trained models, the pairing we first ran. Under every pairing post-training significantly raises top-1 in at most two of six pairs. With the same examples these are Qwen3-8B (+13.7) and Mistral (+10.8); Qwen2.5-7B gains +1.8 and the other three lose 4.8–5.8 points, all n.s.

With reference-free base models only Qwen3-8B gains (+10.7), while Qwen2.5-32B (−8.0) and OLMo 3 7B (−14.5) lose significantly. The example lists matter little to the base models (top-1 within 2.7 points, except OLMo 3.1 32B, 0.630 against 0.559) but raise five of the six post-trained models (Mistral 0.417 to 0.577, OLMo 3 7B 0.258 to 0.345; Qwen2.5-7B-Instruct falls to 0.458). Under every pairing the two smaller Qwen models' lists contain the true pathology more often (recall@5 +16.8 to +26.0) and name more warranted severe conditions, every Qwen list is closer to the reference, and OLMo 3 7B's lists are farther. Together with the few-shot comparison, where the sampled pass@1 gain is already only 1.6–5.2 points in four of six pairs, most of post-training's advantage at one sampled call for the 7–8B Qwen models (+12.6 to +27.5 points at T=0.7 and 1, §4.1) is therefore prompt format and the base model's sampling noise; Qwen3-8B and Mistral keep a real gain.

Temperature frontier. With worked examples and top-p fixed at 1, we sampled the three 7–8B pairs at $T = 0.4$, 0.7, 1.0 and 1.3 (Table 20). Post-trained pass@1 barely moves with temperature (Qwen3-8B 0.60, Qwen2.5-7B-Instruct 0.43–0.46, OLMo-3-7B-Instruct 0.29–0.35), pass@32 rises (to 0.70, 0.66 and 0.74 at T=1.3), and the stated list barely changes (TV 0.68, 0.70 and 0.81–0.83), consistent with the concentration described by Yang et al. (2026). Base models trade pass@1 (Qwen3-8B-Base 0.45 to 0.18) for pass@32, which peaks at T=1 or 1.3 (0.85–0.88), and 25–36% of their answers are invalid at T=1.3. The frontiers differ by family. For Qwen3-8B the post-trained model is better at pass@1 and the base model at pass@32 at every temperature tested; for Qwen2.5-7B the curves meet (at pass@1 about 0.43, the base model at T=0.4 and the Instruct model at T=1.3 both reach pass@32 0.66); and for OLMo 3 7B the base model at T=0.4 beats its Instruct model at every temperature on both (0.368 and 0.803, against at best 0.345 and 0.737). Agreement becomes a better error signal for post-trained models as temperature rises (Qwen3-8B 0.57 to 0.68) but stays below the base models' (0.81–0.92).

Must-not-miss prompt. The standard prompt asks for the most likely conditions. We asked the six post-trained models for the five most likely conditions and then up to three dangerous conditions to rule out (Table 21). Whole lists then name 0.44–0.58 of the severe conditions that the reference weights at least 0.1, 6–15 points more than with the standard prompt and 11–21 more than the base models' lists, and acute pulmonary edema is recovered for every model but Qwen3-8B (e.g. Qwen2.5-32B 0.44 → 0.63, OLMo 3 7B 0.31 → 0.58). The cost is 1.1–2.1 more unwarranted severe conditions per list, mostly pulmonary embolism, named in 88–100% of eligible cases, and for Qwen3-8B, Qwen2.5-7B-Instruct and Mistral a shorter most-likely part that contains the true pathology 5–10 points less often. Even when asked, no model names more than 0.58 of the warranted severe conditions.

F.4. Must-not-miss conditions defined outside DDXPlus

The severity label used above is the dataset's own. We therefore re-ran every severe-condition analysis with two published sets of conditions that must not be missed, mapped onto DDXPlus's 49 (Table 22): a harm-based set, the DDXPlus conditions among the 15 diseases that cause about half of serious misdiagnosis-related harms in the USA (Newman-Toker et al., 2024) (NSTEMI/STEMI, pulmonary embolism, pneumonia and pulmonary neoplasm, of which DDXPlus labels only the first two severe), and the life-threatening

Pair	Top-1 accuracy base, reference	base, ref.-free	post-trained zero-shot	post-trained examples	Top-1, post − base (points) same examples	ref.-free base	original
Qwen3-8B	0.471	0.450	0.558	0.607	+13.7	+10.7	+8.7†
Qwen2.5-7B	0.437	0.420	0.496	0.458	+1.8†	+7.6†	+5.7†
Qwen2.5-32B	0.712	0.686	0.606	0.664	−4.8†	−8.0	−10.6
OLMo 3 7B	0.403	0.403	0.258	0.345	−5.8†	−14.5	−14.5
OLMo 3.1 32B	0.630	0.559	0.558	0.571	−5.7†	−0.1†	−7.1†
Mistral-7B	0.468	0.446	0.417	0.577	+10.8	−2.9†	−5.1†

Pair	Recall@5, post − base (points) same	ref.-free	original	TV one list, post − base same	ref.-free	original	Severe named, post − base same	ref.-free	original
Qwen3-8B	+26.0	+21.6	+24.2	−0.108	−0.077	−0.088	+0.127	+0.127	+0.127
Qwen2.5-7B	+20.1	+18.3	+16.8	−0.074	−0.075	−0.060	+0.091	+0.119	+0.110
Qwen2.5-32B	+0.3†	−2.4†	−5.3†	−0.054	−0.035	−0.031	+0.058	+0.046	+0.062
OLMo 3 7B	+2.0†	−17.2	−14.1	+0.060	+0.067	+0.083	+0.004†	−0.058†	−0.020†
OLMo 3.1 32B	+2.5†	−0.5†	−2.0†	+0.004†	+0.004†	+0.012†	−0.005†	−0.033†	−0.042†
Mistral-7B	+16.7	+2.9†	+4.1†	−0.009†	+0.008†	+0.029	−0.034†	−0.034†	−0.059†

Table 19: Greedy single call (DDXPlus, 980 cases, temperature 0, one answer per case). Base models are prompted with the worked examples of §F.1, whose lists are the reference top five ("reference") or Qwen2.5-32B-Instruct's answers ("ref.-free"); post-trained models are prompted zero-shot or with the reference examples. Three comparisons: *same examples* (both arms see the reference examples), *ref.-free base* (reference-free base against zero-shot post-trained, so neither arm sees a reference list), and *original* (reference-example base against zero-shot post-trained). Differences are over cases where both models gave a valid answer, so they can differ slightly from the difference of the level columns. †: the condition-clustered 95% interval includes zero.

causes of acute chest pain in the 2021 AHA/ACC guideline (Gulati et al., 2021) (acute coronary syndrome as NSTEMI/STEMI and unstable angina, pulmonary embolism, esophageal rupture as Boerhaave, and pneumothorax). Their union has seven conditions. Neither set involves a judgment of these vignettes, and both were fixed before the analysis. On average the result is stronger than with DDXPlus's labels. Zero-shot, post-trained lists name 9–24 points more of the warranted union conditions in seven of eight settings (OLMo 3.1 32B −0.3, n.s.), and with worked examples 7–22 points more for the three Qwen models and OLMo 3 7B (OLMo 3.1 32B +6.5 and Mistral +4.0, n.s.); the harm-based and chest-pain sets give the same zero-shot pattern. The losses of specific conditions, however, are confined to OLMo, Mistral and the few-shot Qwen2.5 models. No Qwen model names any union condition significantly less often zero-shot, while OLMo 3 7B loses NSTEMI/STEMI and unstable angina at both temperatures (and pulmonary neoplasm at T=1), OLMo 3.1 32B loses NSTEMI/STEMI, pulmonary embolism and pneumothorax, and with worked examples Mistral loses four of the seven (pulmonary embolism 31 points), Qwen2.5-7B-Instruct two and Qwen2.5-32B-Instruct one (Boerhaave). When the true pathology is itself in the union set (140 cases), zero-shot post-trained lists contain it among their top five 10–26

Model	T	Invalid (%)	pass@1	pass@32	Distinct	TV one list	AUROC stated	AUROC agree@32
Qwen3-8B-Base	0.4	0.1	0.447	0.727	3.64	0.772	0.687	0.806
	0.7	0.3	0.388	0.832	7.09	0.774	0.708	0.876
	1.0	3.0	0.304	0.880	10.33	0.800	0.703	0.909
	1.3	29.5	0.179	0.857	10.39	0.850	0.630	0.921
Qwen3-8B	0.4	0.0	0.602	0.632	1.14	0.681	0.693	0.569
	0.7	0.0	0.601	0.655	1.25	0.682	0.693	0.613
	1.0	0.0	0.599	0.676	1.40	0.682	0.695	0.651
	1.3	0.0	0.596	0.703	1.57	0.682	0.694	0.679
Qwen2.5-7B	0.4	0.4	0.421	0.662	3.13	0.778	0.748	0.805
	0.7	0.6	0.388	0.784	5.72	0.790	0.719	0.843
	1.0	2.6	0.333	0.841	8.53	0.815	0.692	0.899
	1.3	25.3	0.220	0.848	9.24	0.857	0.615	0.916
Qwen2.5-7B-Instruct	0.4	0.0	0.455	0.516	1.40	0.702	0.750	0.613
	0.7	0.0	0.448	0.556	1.78	0.703	0.747	0.643
	1.0	0.0	0.441	0.597	2.28	0.704	0.741	0.687
	1.3	0.2	0.430	0.660	2.98	0.707	0.720	0.729
OLMo-3-1025-7B	0.4	0.1	0.368	0.803	5.46	0.762	0.667	0.815
	0.7	0.4	0.294	0.865	9.62	0.795	0.663	0.879
	1.0	2.8	0.214	0.881	12.40	0.834	0.618	0.899
	1.3	36.2	0.110	0.801	10.69	0.888	0.552	0.882
OLMo-3-7B-Instruct	0.4	0.1	0.345	0.504	1.99	0.809	0.737	0.663
	0.7	0.1	0.335	0.602	2.95	0.808	0.740	0.716
	1.0	0.1	0.324	0.668	4.09	0.809	0.744	0.760
	1.3	5.6	0.286	0.737	5.88	0.831	0.680	0.775

Table 20: Temperature frontier with the few-shot prompt (DDXPlus, 980 cases, 32 samples, top-p 1 at every temperature).

points more often for four of five pairs (OLMo 3 7B +9.9, n.s.). Scored against the union set, the must-not-miss prompt names 15–26 points more of the warranted conditions than the standard prompt and 8–20 points more than two combined standard lists, a larger share of its gain than under DDXPlus's labels, at a cost of 0.7–1.5 more unwarranted union conditions per list.

F.5. Chance-corrected pass@32 and other models' views

Chance. With 49 listed conditions, part of pass@32 comes from answers that do not depend on the patient. We therefore also score each case's samples against the true pathology of a random case with a different pathology (Table 23). Base models' samples then still reach 0.05–0.19 at pass@32 and post-trained ones 0.01–0.04, far below the 0.48 of a uniform guesser, because both concentrate on conditions that fit the presentation. Subtracting these values shrinks the loss but keeps it in every pair with worked examples: 6.7–17.8 points with the reference-derived examples and 4.1–14.3 with another model's. Zero-shot, the cor-

Model	Valid (%)	Conditions likely + mnm	Recall@5 default	Recall@5 likely part	Severe named (ref. ≥ 0.1) base	Severe named (ref. ≥ 0.1) default	Severe named (ref. ≥ 0.1) mnm	Unwarranted severe default	Unwarranted severe mnm	Acute pulm. edema default	Acute pulm. edema mnm
Qwen3-8B	100	4.1 + 2.3	0.802	0.715	0.318	0.437	0.525	1.06	2.11	0.61	0.60
Qwen2.5-7B-Instruct	99	3.9 + 2.9	0.735	0.640	0.304	0.439	0.503	1.57	2.95	0.63	0.71
Qwen2.5-32B-Instruct	100	4.8 + 2.9	0.840	0.838	0.414	0.472	0.579	0.91	2.27	0.44	0.63
OLMo-3-7B-Instruct	96	3.7 + 2.5	0.485	0.501	0.332	0.317	0.444	1.15	2.60	0.31	0.58
OLMo-3.1-32B-Instruct	100	4.9 + 3.0	0.779	0.790	0.376	0.392	0.545	0.74	2.85	0.41	0.59
Mistral-7B-Instruct	87	3.7 + 2.7	0.613	0.562	0.352	0.329	0.473	0.86	2.29	0.59	0.68

Table 21: Must-not-miss prompt (DDXPlus, T=0.7, 980 cases, 32 samples): the five most likely conditions, then up to three dangerous conditions to rule out. *Default*: the same model with the standard zero-shot prompt; *base*: its base model with the few-shot prompt. Recall@5 of the most-likely part; severe conditions and unwarranted severe conditions (severe, reference < 0.1, per list) over the whole list; acute pulmonary edema: share of lists naming it where the reference gives it any weight.

rected loss is 7.5 and 3.6 points for the 7–8B Qwen models and 25.2 for OLMo 3 7B, and it disappears at 32B (−0.5 and +0.1).

Other models' views. The packing test of Table 27 compares a post-trained list with its base model's mean for the same patient against the base model's means for other patients with the same diagnosis, a lenient baseline that any list responding to the patient passes. Against other families' base models alone, every zero-shot post-trained list is closer to its own base model's mean (−0.031 to −0.138). Table 3 compares it with every other final model's mean for the same patient. For Qwen3-8B, Qwen2.5-32B and OLMo 3.1 32B the post-trained list is closer to its own base model's view than to any other base model's (by 0.036–0.058) and far closer than to the average post-trained model's (0.067–0.090), but about as close to the nearest other post-trained model's (n.s.). Qwen2.5-7B-Instruct is as close to Qwen3-8B-Base's view as to its own base model's, and OLMo-3-7B-Instruct is closer to Qwen3-8B-Base's and Qwen3-8B's. For the stronger models, then, a post-trained list stays close to its own base model's view, while the stronger post-trained models also converge on one another.

F.6. Ties, list size, validity, prevalence and the reference ceiling

These checks use the zero-shot T=0.7 final pairs. *Ties.* Post-trained models' 32 samples are unanimous in 34–57% of cases (Qwen3-8B 78%), where agreement cannot rank anything. On the remaining cases agreement's AUROC is 0.61–0.66 for the post-trained models against 0.78–0.83 for their base models, so agreement weakens even where it is not tied. On those cases Qwen3-8B's stated probability no longer beats agreement (0.611 against 0.617): its advantage comes from cases where all samples agree, whereas OLMo-3-7B-Instruct's stated probability stays ahead (0.729 against 0.605). *List size.* The three most frequent of 32 samples can be fewer than three answers. With $m = \min(3, \text{distinct sampled answers})$, the first m items of one list contain the true pathology about as often as the m most frequent samples for the post-trained models (−0.5 to −5.2 points; significantly lower for Qwen2.5-32B-Instruct and OLMo-3.1-32B-Instruct), while for base models the samples win by 8.5–14.6. The advantage of a post-trained list's top three is that it names three conditions where

Pair	Union set (7 conditions) base	post	post − base	sig. lost / gained	Harm-based post − base	Chest pain post − base
Qwen3-8B, T=0.7	0.494	0.584	+0.089	0 / 2	+0.053	+0.115
Qwen3-8B, T=1	0.425	0.585	+0.160	0 / 4	+0.135	+0.167
Qwen2.5-7B, T=0.7	0.315	0.536	+0.221	0 / 4	+0.267	+0.149
Qwen2.5-7B, T=1	0.303	0.539	+0.236	0 / 4	+0.287	+0.180
Qwen2.5-32B, T=0.7	0.432	0.595	+0.162	0 / 4	+0.164	+0.104
OLMo 3 7B, T=0.7	0.268	0.413	+0.144	2 / 3	+0.133	+0.101
OLMo 3 7B, T=1	0.275	0.412	+0.136	3 / 3	+0.135	+0.102
OLMo 3.1 32B, T=0.7	0.462	0.459	−0.003†	3 / 0	−0.033†	−0.020†
Qwen3-8B, few-shot	0.299	0.491	+0.191	0 / 3	+0.183	+0.093
Qwen2.5-7B, few-shot	0.239	0.455	+0.216	2 / 3	+0.249	+0.152
Qwen2.5-32B, few-shot	0.361	0.542	+0.180	1 / 3	+0.210	+0.096
OLMo 3 7B, few-shot	0.335	0.409	+0.074	2 / 3	−0.015†	+0.036†
OLMo 3.1 32B, few-shot	0.412	0.477	+0.065†	2 / 1	+0.080	−0.017†
Mistral-7B, few-shot	0.401	0.441	+0.040†	4 / 1	+0.024†	−0.071

Post-trained model	Warranted must-not-miss conditions named (union set) standard	two lists	must-not-miss	− standard	Unwarranted per list standard	two lists	must-not-miss
Qwen3-8B	0.584	0.612	0.775	+0.192	1.08	1.20	2.10
Qwen2.5-7B-Instruct	0.536	0.600	0.714	+0.178	0.94	1.14	2.33
Qwen2.5-32B-Instruct	0.595	0.648	0.814	+0.219	1.00	1.21	2.47
OLMo-3-7B-Instruct	0.413	0.479	0.564	+0.151	0.93	1.17	1.60
OLMo-3.1-32B-Instruct	0.459	0.527	0.715	+0.256	0.58	0.77	1.77
Mistral-7B-Instruct	0.358	0.417	0.615	+0.257	0.69	0.87	1.65

Table 22: Must-not-miss conditions defined outside DDXPlus (DDXPlus, 980 cases). *Harm-based*: the DDXPlus conditions among the 15 diseases behind about half of serious misdiagnosis-related harms in the USA (Newman-Toker et al., 2024) (NSTEMI/STEMI, pulmonary embolism, pneumonia, pulmonary neoplasm). *Chest pain*: the life-threatening causes of acute chest pain in the 2021 AHA/ACC guideline (Gulati et al., 2021) (NSTEMI/STEMI, unstable angina, pulmonary embolism, Boerhaave, spontaneous pneumothorax). *Union*: either set. Top: share of the set's conditions with reference probability ≥ 0.1 that one list names, and the number of the union's conditions named significantly less / more often (Benjamini–Hochberg over the 7 conditions). Bottom: the must-not-miss prompt (§F.3) scored against the union set; *two lists*: the union of two standard lists, a length-matched comparison; unwarranted: the set's conditions named that the reference weights below 0.1. †: the condition-clustered 95% interval includes zero.

its samples offer fewer. *Validity.* Counting invalid samples as wrong (as everywhere in the paper) or scoring valid samples only leaves pass@32 unchanged and moves pass@1, mostly for OLMo 3 7B, whose single-answer gain is +7.1 points with invalid samples counted as wrong and −3.5 (n.s.) over valid samples only, weighted to the natural prevalence below. *Prevalence.* The test sample is class-balanced (20 cases per pathology). Reweighted to the pathology frequencies of the full DDXPlus test split (134,529 patients), post-training costs 4.9–31.5 points of pass@32 (all intervals exclude zero, including both 32B pairs), raises recall@5 by 10.3–17.7 points for the Qwen models and OLMo 3.1 32B (−0.3 for OLMo 3

Pair	Zero-shot shuffled (base / post)	Zero-shot raw	Zero-shot corrected	Worked examples (reference lists) shuffled (base / post)	Worked examples (reference lists) raw	Worked examples (reference lists) corrected	Worked examples (another model's lists) shuffled (base / post)	Worked examples (another model's lists) raw	Worked examples (another model's lists) corrected
Qwen3-8B	0.063 / 0.014	−12.4	−7.5	0.104 / 0.012	−17.1	−7.9	0.072 / 0.010	−14.9	−8.7
Qwen2.5-7B	0.088 / 0.026	−9.8	−3.6	0.087 / 0.022	−22.9	−16.4	0.080 / 0.022	−20.1	−14.3
Qwen2.5-32B	0.063 / 0.018	−4.9	−0.5	0.056 / 0.015	−15.3	−11.2	0.049 / 0.014	−14.5	−10.9
OLMo 3 7B	0.088 / 0.039	−30.1	−25.2	0.163 / 0.041	−30.0	−17.8	0.191 / 0.034	−29.9	−14.3
OLMo 3.1 32B	0.072 / 0.029	−4.3	+0.1	0.083 / 0.020	−19.1	−12.7	0.095 / 0.020	−17.6	−10.0
Mistral-7B	—	—	—	0.091 / 0.017	−14.1	−6.7	0.097 / 0.017	−12.1	−4.1

Table 23: Chance-corrected pass@32 (DDXPlus, T=0.7, 980 cases). *Shuffled*: pass@32 when each case's samples are scored against the true pathology of a random case with a different pathology (mean of 20 permutations), the part of pass@32 that does not depend on the patient. *Raw*: post-trained minus base pass@32, in points; *corrected*: the same after subtracting each model's shuffled value. Mistral's base model does not answer the zero-shot prompt (§F.8). This check takes a sample's top answer to be its first listed name that maps to one of the 49 conditions, so raw values can differ from Tables 6 and 16 by up to 1.8 points.

7B) and lowers one list's TV by 0.047–0.069 (OLMo 3 7B +0.031). *Reference ceiling.* The reference's own top five contains the true pathology in 96.9% of cases (97.3% at natural prevalence), against 0.47–0.84 for one model list. *What replaces acute pulmonary edema.* On the 54 cases whose reference gives acute pulmonary edema at least 0.1, the conditions whose share of lists rises most after post-training are mostly other must-not-miss conditions (pulmonary embolism for Qwen2.5-7B-Instruct and OLMo-3-7B-Instruct, +0.29 and +0.52; pneumonia and pulmonary embolism for Qwen2.5-32B-Instruct; unstable angina, panic attack and NSTEMI/STEMI for Qwen3-8B), but atrial fibrillation and influenza for OLMo-3.1-32B-Instruct, whose lists name acute pulmonary edema 0.25 less often.

F.7. List concentration, dangerous misses, selective prediction and common cases

Table 24 adds four checks computed with the same loader as the headline tables. *Concentration.* A five-item list could not get shorter, but its probabilities could concentrate. Zero-shot, the entropy of one list changes by −0.13 to +0.08 nats for the Qwen models and OLMo 3.1 32B, and with worked examples it rises for them (+0.05 to +0.25); only OLMo-3-7B-Instruct (−0.52 zero-shot) and Mistral-Instruct (−0.18 with examples) state markedly more concentrated lists. *Dangerous misses.* Per case, post-trained lists leave out a severe true pathology 9–18 points less often zero-shot and name none of the warranted severe conditions about 10 points less often (OLMo 3.1 32B −2.2, n.s.). *Selective prediction.* Ranking cases by agreement gives the higher accuracy at 50% coverage for every base model. After post-training the stated probability does better for Qwen3-8B and OLMo-3-7B-Instruct zero-shot and for every post-trained model with worked examples; the area under the risk–coverage curve gives the same ordering. *Common cases.* On the cases where both models give at least 16 valid samples, pass@16 over valid samples falls by 3.4–30.5 points zero-shot (n.s. for Qwen2.5-32B) and distinct answers by 1.5–2.7, so the collapse does not come from invalid answers; and on the cases where neither model's samples are unanimous, agreement's AUROC falls in every zero-shot pair (0.72–0.77 → 0.61–0.67).

Pair	Entropy of one list (nats)	Lists that miss (share) severe true dx	Lists that miss (share) all warranted severe	Accuracy at 50% coverage, stated p / agreement base	Accuracy at 50% coverage, stated p / agreement post-trained	Agreement AUROC, both non-unanimous (n)
Qwen3-8B, zero-shot	1.29 → 1.16	0.38 → 0.20	0.50 → 0.39	0.52 / 0.69	0.74 / 0.68	0.74 → 0.62 (214)
Qwen2.5-7B, zero-shot	1.41 → 1.35	0.39 → 0.28	0.49 → 0.39	0.41 / 0.60	0.54 / 0.65	0.72 → 0.61 (477)
Qwen2.5-32B, zero-shot	1.32 → 1.40	0.31 → 0.16	0.46 → 0.35	0.68 / 0.75	0.74 / 0.77	0.77 → 0.64 (417)
OLMo 3 7B, zero-shot	1.41 → 0.89	0.55 → 0.46	0.59 → 0.50	0.31 / 0.46	0.44 / 0.41	0.77 → 0.61 (646)
OLMo 3.1 32B, zero-shot	1.27 → 1.33	0.37 → 0.25	0.47 → 0.45	0.53 / 0.70	0.70 / 0.79	0.73 → 0.67 (544)
Qwen3-8B, examples	1.29 → 1.34	0.37 → 0.15	0.54 → 0.40	0.55 / 0.68	0.74 / 0.63	0.66 → 0.53 (136)
Qwen2.5-7B, examples	1.16 → 1.33	0.38 → 0.21	0.55 → 0.41	0.55 / 0.62	0.65 / 0.53	0.73 → 0.60 (320)
Qwen2.5-32B, examples	1.20 → 1.43	0.12 → 0.10	0.42 → 0.38	0.79 / 0.89	0.81 / 0.78	0.84 → 0.55 (228)
OLMo 3 7B, examples	1.26 → 0.90	0.43 → 0.38	0.51 → 0.50	0.41 / 0.55	0.51 / 0.47	0.81 → 0.58 (648)
OLMo 3.1 32B, examples	1.12 → 1.37	0.27 → 0.20	0.48 → 0.42	0.67 / 0.83	0.77 / 0.76	0.72 → 0.62 (351)
Mistral-7B, examples	0.93 → 0.75	0.38 → 0.25	0.49 → 0.49	0.56 / 0.70	0.76 / 0.65	0.75 → 0.67 (236)

Table 24: Further checks (DDXPlus, T=0.7, base → post-trained). *Entropy*: of one list's renormalized probabilities, per case mean. *Lists that miss*: share of one model's lists that do not contain a severe true pathology among their five items (cases whose true pathology has severity 1–2), and that name none of the severe conditions the reference weights at least 0.1 (cases with at least one). *Accuracy at 50% coverage*: accuracy of the single-call answer on the half of cases ranked most confident by the stated probability or by 31-sample agreement. *Agreement AUROC*: on the cases where neither model's valid samples are unanimous, so both models are compared on the same cases and agreement is not tied.

F.8. A third family: Mistral-7B v0.3

We repeat the comparison with Mistral-7B-v0.3 and Mistral-7B-Instruct-v0.3, a family also studied by Jang et al. (2026). With the zero-shot prompt the base model does not answer: 52% of its samples at T=0.7 and 76% at T=1 are invalid (it continues the document instead), and its valid answers mostly repeat the first listed condition, so its single-call accuracy is 6% and the zero-shot pair is not informative. With the few-shot prompt of §F.1 the base model answers validly (0.4% invalid), and the pair reproduces the other families (Table 25). Post-training raises pass@1 by 14.0 points and lowers pass@32 by 14.5, cuts distinct answers from 5.0 to 1.3 and Cov@32 by 0.138, and moves the samples away from the reference (TV +0.058). Voting gains the base model 5.6 points and the Instruct model 0.8; the three most frequent of 32 samples beat one list by 12.6 points for the base model, while for the Instruct model the list is ahead by 7.0; agreement is the better error signal for the base model (AUROC 0.826 against 0.671) and the stated probability for the Instruct model (0.747 against 0.613). The stated list improves modestly (TV −0.036 and DDF1 +0.046, surviving clustering and Holm correction over 42 tests; −0.023 with the true pathology removed, which does not), the smallest significant gain of the three families, and 11 of the 17 severe conditions are named less often (3 significantly). Mistral-Instruct also names 0.5 conditions per list that are not among the 49.

Appendix G. Thinking mode, aggregation and the error signal

G.1. Thinking mode

Qwen3-8B can reason in a thinking block before it answers; the main runs turn this off. With thinking on (zero-shot prompt, T=0.7, up to 8,192 tokens; traces average about

1,500 tokens and all end before the limit), the same weights behave differently (Table 25). Accuracy rises at both ends: pass@1 is 0.671 against 0.559 without thinking, and pass@32 is 0.815, above the non-thinking model (0.607) and level with the base model (0.732 zero-shot, +8.4 [1.7, 15.4] points, not surviving Holm correction over 42 tests; 0.809 with worked examples). The samples are still narrow (1.76 distinct answers against 3.77 for the base model; Cov@32 0.283 against 0.323) but no farther from the reference than the base model's (TV 0.745 against 0.750). The stated list is the closest of any 7–8B model (TV 0.670, against 0.700 without thinking). The shift in error signals reverses: agreement across 31 samples is again the better signal (AUROC 0.732 against 0.638; difference 0.094 [0.027, 0.170]), as for base models, and voting gains 1.5 points (n.s.). The narrowing of the samples and the advantage of the stated probability are therefore properties of answering directly, which a reasoning mode largely undoes, consistent with the chain-of-thought prompt of §D.3.

G.2. Averaging stated lists across calls

Stated lists can be averaged across calls, as samples can be voted. For each case we average the first 5 or 32 valid stated lists, each renormalized over the conditions it names, and score the mean against the reference (Table 26). Averaging helps base models far more than post-trained ones, whose lists vary little between calls (at T=0.7 with the zero-shot prompt, 32 averaged lists name 7.6–13.4 conditions, against 16.6–22.0 for base models). The mean of five base-model lists is about as close to the reference as one post-trained list (Qwen3-8B 0.699 against 0.700, Qwen2.5-7B 0.717 against 0.717, OLMo 3.1 32B 0.691 against 0.701). With five lists on both sides, the post-trained mean is closer for Qwen2.5-7B under both prompts (-0.025 and -0.027) and for Qwen2.5-32B and Qwen3-8B under one prompt each, not significantly closer for OLMo 3.1 32B (zero-shot) and Mistral (few-shot), and farther for OLMo 3 7B and few-shot OLMo 3.1 32B. Averaged base lists also name severe conditions at least as often, and more often for OLMo 3.1 32B, OLMo 3 7B, Mistral and few-shot Qwen2.5-32B (by 0.06–0.14). One post-trained call thus gets about as close to the reference as several base calls; the packing test (Table 27) shows that it is a stabilized version of the base model's view of the patient rather than its literal average, and, as for the open-ended questions of Lake et al. (2025), post-training adds little that averaged base lists lack (five post-trained lists beat five base lists in four of eleven settings). Qwen3-8B's thinking mode goes furthest: five of its lists beat five base-model lists by more than any direct-answer model's (TV -0.049) and name more severe conditions (+0.129). For a post-trained model answering directly, averaging five lists instead of reading one improves TV by only 0.005–0.025 at T=0.7 (up to 0.038 at T=1); it names 6–17 points more severe conditions, mostly by naming 1.4 to 3.6 more conditions. Five post-trained lists still cover 0.44–0.50 of the reference mass at T=0.7 (OLMo 3 7B 0.37) with six to eight conditions, about twice what their 32 samples cover (0.22–0.28) and as much as a uniform guesser naming about 24 conditions (0.47), so the coverage deficit of §4.1 is one of the sampled readout.

G.3. Which signal to use: combination and held-out audit

Whether the stated probability or agreement across samples is the better error signal varies across models, so we ask what predicts it and whether a practitioner can tell. We pool 56 model–setting combinations scored on DDXPlus: the base/post-trained pairs at both

Model	Invalid (%)	pass@1	pass@32	Distinct	Cov@32	TV, one list all	TV, one list w/o true	DDF1	TV samples	Vote@32 gain	Top-3 list − samples	AUROC stated	AUROC agree@5
Mistral-7B-v0.3 (base), few-shot	0.4	0.428	**0.791**	**5.02**	**0.355**	0.777	0.875	0.250	**0.741**	+5.6	−12.6	0.671	0.826
Mistral-7B-Instruct-v0.3, few-shot	0.3	**0.568**	0.646	1.34	0.217	**0.741**	**0.851**	**0.296**	0.799	+0.8	+7.0	0.747	0.613
Qwen3-8B-Base	11.3	0.388	0.732	**3.77**	**0.323**	0.743	0.833	0.276	0.750	+3.4	−7.3	0.595	0.772
Qwen3-8B, non-thinking	0.0	0.559	0.607	1.27	0.217	0.700	0.815	0.318	0.793	+0.3	+13.7	0.734	0.613
Qwen3-8B, thinking	0.0	**0.671**	**0.815**	1.76	0.283	**0.670**	**0.799**	**0.349**	**0.745**	+1.5	+3.0	0.638	0.693

Table 25: A third family and a reasoning mode on DDXPlus (T=0.7, 980 cases, 32 samples). Top: Mistral-7B v0.3 with the few-shot prompt of §F.1. Bottom: Qwen3-8B-Base and Qwen3-8B with thinking off (the main runs) and on, with the zero-shot prompt. Columns as in Table 15; Distinct: distinct top answers per 32 samples. Bold: best within each block.

Pair	TV, base model 1 list	TV, base model mean of 5	TV, base model mean of 32	TV, post-trained model 1 list	TV, post-trained model mean of 5	TV, post-trained model mean of 32	Post − base, 5 lists each TV	Post − base, 5 lists each Severe named
Zero-shot prompt								
Qwen3-8B	0.743	0.699	0.691	0.700	0.695	0.692	−0.004†	+0.030†
Qwen2.5-7B	0.779	0.717	0.706	0.717	0.692	0.685	−0.025	+0.023†
Qwen2.5-32B	0.726	0.681	0.674	0.674	0.658	0.654	−0.024	−0.037†
OLMo 3 7B	0.794	0.730	0.717	0.823	0.805	0.801	+0.074	−0.076
OLMo 3.1 32B	0.746	0.691	0.678	0.701	0.677	0.673	−0.014†	−0.088
Few-shot prompt								
Qwen3-8B	0.768	0.704	0.689	0.682	0.673	0.672	−0.030	−0.006†
Qwen2.5-7B	0.784	0.712	0.694	0.702	0.686	0.683	−0.027	+0.012†
Qwen2.5-32B	0.712	0.655	0.641	0.654	0.640	0.638	−0.015†	−0.094
OLMo 3 7B	0.787	0.703	0.676	0.808	0.786	0.778	+0.083	−0.141
OLMo 3.1 32B	0.719	0.651	0.630	0.697	0.682	0.680	+0.031	−0.060
Mistral-7B v0.3	0.777	0.708	0.691	0.741	0.723	0.720	+0.015†	−0.118
Zero-shot prompt, thinking on								
Qwen3-8B, thinking	0.743	0.699	0.691	0.670	0.650	0.647	−0.049	+0.129

Table 26: Averaging stated lists across calls (DDXPlus, T=0.7, 980 cases). TV to the reference differential of one stated list (per-case mean over valid lists, as in Table 2) and of the mean of the first 5 or 32 valid lists, for the base and the final post-trained model. Last two columns: post-trained minus base with five averaged lists each, for TV and for the share of severe conditions with reference probability ≥ 0.1 that are named. †: the condition-clustered 95% interval includes zero.

temperatures and with three prompts, the OLMo stages, the 32B models, the medical models (including three fine-tuned from Llama-3.1-8B, HuatuoGPT-o1-8B, AlphaMed-8B and UltraMedical-8B, which appear only here), our GRPO runs, and the full-distribution prompt. They share weights, prompts and temperatures, so we treat the correlations below as descriptive. The stated probability's advantage over five-sample agreement correlates with the *single-answer rate*, the share of cases whose five samples all give the same top answer (Spearman 0.75), with the stated probability's own AUROC (0.82; the gap contains this term, so part of this is by construction), and negatively with the number of distinct answers (−0.74). The stated probability is the better signal for 19 of the 56. No single-answer

Pair	TV of a post-trained list to own base mean	same-diagnosis means	reference	Within-model benchmark base list	post list	Differences own − same dx	own − base list
Zero-shot prompt							
Qwen3-8B, T=0.7	0.368	0.444	0.701	0.372	0.109	−0.076	−0.004†
Qwen3-8B, T=1	0.424	0.486	0.701	0.551	0.153	−0.062	−0.128
Qwen2.5-7B, T=0.7	0.464	0.519	0.717	0.470	0.248	−0.055	−0.006†
Qwen2.5-7B, T=1	0.532	0.575	0.717	0.650	0.315	−0.043	−0.119
Qwen2.5-32B, T=0.7	0.360	0.431	0.674	0.375	0.189	−0.071	−0.015†
OLMo 3 7B, T=0.7	0.628	0.658	0.823	0.493	0.238	−0.031	+0.135
OLMo 3 7B, T=1	0.651	0.679	0.818	0.691	0.326	−0.028	−0.040
OLMo 3.1 32B, T=0.7	0.386	0.452	0.700	0.432	0.227	−0.066	−0.046
Few-shot prompt							
Qwen3-8B	0.469	0.518	0.681	0.471	0.102	−0.049	−0.003†
Qwen2.5-7B	0.559	0.596	0.702	0.515	0.171	−0.037	+0.044
Qwen2.5-32B	0.465	0.511	0.654	0.441	0.154	−0.045	+0.024
OLMo 3 7B	0.669	0.710	0.808	0.594	0.259	−0.040	+0.075
OLMo 3.1 32B	0.482	0.532	0.697	0.469	0.150	−0.050	+0.014†
Mistral-7B	0.495	0.546	0.740	0.495	0.181	−0.051	−0.000†
Thinking on							
Qwen3-8B, thinking	0.412	0.468	0.669	0.372	0.189	−0.057	+0.040

Table 27: Does a post-trained list reproduce its base model's view of the patient? (DDXPlus, 980 cases.) TV between a post-trained list and the mean of the base model's valid lists for the same case (own base mean), the base model's means for the other cases with the same true pathology, and the reference. Within-model benchmarks: a base list against the mean of the base model's other lists for the case, and a post-trained list against the mean of the post-trained model's other lists. †: the condition-clustered 95% interval includes zero.

Model (T=0.7)	Stated p	Agree@5	Agree@32
Qwen3-8B-Base	+0.042	+0.005	+0.024
Qwen3-8B	+0.116	+0.056	+0.098
Qwen2.5-7B	+0.044	+0.025	+0.038
Qwen2.5-7B-Instruct	+0.070	+0.054	+0.070
Qwen2.5-32B	+0.064	−0.026	−0.038
Qwen2.5-32B-Instruct	−0.043	−0.060	−0.109
OLMo-3-1025-7B	+0.083	−0.048	−0.052
OLMo-3-7B-Instruct	−0.067	−0.073	−0.049
OLMo-3-1125-32B	−0.005	+0.020	+0.011
OLMo-3.1-32B-Instruct	+0.063	+0.014	−0.023

Table 28: Spearman correlation between each uncertainty signal (lower confidence or agreement) and the entropy of the case's reference differential (DDXPlus, 980 cases). Neither readout tracks how ambiguous a case is.

threshold separates them: Qwen2.5-7B-Instruct still favors agreement, while OLMo 3 7B's DPO and Instruct models favor their unusually informative stated probability. With five-sample agreement, as in the original audit, combining the signals beats choosing between them; with agreement across all 31 further samples the advantage shrinks (post-trained combinations: agreement 0.742, rank average 0.761, +0.019 $[-0.007, 0.040]$; logistic 0.772,

Model	Single-call accuracy	Vote@32 gain	Top-3: list − 32 samples	AUROC, single-call answer correct		
				Stated p	Agree@5	Agree@32
Qwen3-8B-Base	0.456	+3.4	−7.3	0.595	0.772	**0.810**
Qwen3-8B	0.556	+0.3	+13.7	**0.734**	0.613	0.655
Qwen3-8B (CoT prompt)	0.648	+4.5	+0.0	0.718	0.725	**0.778**
Qwen3-8B (thinking)	0.672	+1.5	+3.0	0.638	0.693	**0.732**
Qwen2.5-7B	0.371	+5.2	−8.9	0.576	0.764	**0.811**
Qwen2.5-7B-Instruct	0.480	+0.8	+7.3	0.571	0.653	**0.702**
Qwen2.5-32B	0.510	+7.0	−8.3	0.730	0.759	**0.797**
Qwen2.5-32B-Instruct	0.598	+1.9	+5.6	0.709	0.663	**0.721**
OLMo-3 7B base	0.291	+4.0	−13.6	0.538	0.761	**0.802**
OLMo-3-7B-Instruct	0.267	−0.2	+1.9	**0.796**	0.668	0.731
OLMo-3.1 32B base	0.441	+8.4	−13.2	0.629	0.805	**0.831**
OLMo-3.1-32B-Instruct	0.562	+0.9	−0.6	0.696	0.712	**0.772**
AlphaMed-7B	0.528	+0.9	+0.6	0.673	0.663	**0.715**
HuatuoGPT-o1-7B	0.532	+4.8	−0.4	0.590	0.715	**0.775**
m1-7B	0.495	+3.8	−4.6	0.680	0.736	**0.782**

Table 29: Aggregation and uncertainty on DDXPlus (T=0.7, 980 cases). Single-call accuracy: first valid sample. Vote@32 gain: majority answer over valid samples minus single call (points). Top-3: one list's top three minus the three most frequent sampled answers, containing the true pathology (points). AUROC for the single-call answer being correct; bold marks the best signal.

+0.030 [0.013, 0.044]), and for the 17 base combinations agreement alone (0.821) beats the rank average (0.780) and ties the logistic combination (0.827). The rest of this section uses five-sample agreement. Fitting a logistic model of correctness on the stated probability and five-sample agreement with 100 labeled cases, and evaluating on the other 880, gives a mean held-out AUROC of 0.753 for the 39 post-trained combinations (0.765 for all 56); averaging the two signals' ranks without fitting gives 0.746; picking the better single signal on the same 100 cases gives 0.720; always using agreement 0.695; always using the stated probability 0.671. The logistic combination beats the single-signal pick for 35 of the 39 post-trained combinations. This matches the general finding that hybrids of consistency and verbalized confidence beat either (Xiong et al., 2024). For choosing a single signal, the simplest rule, always using agreement, is right for 66% of the combinations (54% of the 39 post-trained ones) and loses 0.025 AUROC on average (0.032). An audit does better: choose the signal with the higher AUROC on 100 random labeled cases, using five samples per case, and evaluate the choice on the other 880. Over 500 draws per combination it picks the better held-out signal 86% of the time (82% for post-trained models) and loses 0.005 AUROC (0.007); with 50 cases, 82% and 0.010; with 200, 90% and 0.003. It is reliable when the two signals differ by more than 0.05 AUROC (96% of draws, 39 combinations) and close to a coin flip when they differ by less than 0.02 (53%, 5 combinations), where the choice matters least. The audit's main cost is gold labels for 100 cases from the target population, and we validate it only within DDXPlus.

Appendix H. Real cases and medical post-training

H.1. Medical post-training

Applied after instruction tuning, medical RL does not cost coverage. AlphaMed improves on its parent at every k (Figure 4): +3.9 points at k=1 ([+2.2, +5.5]; clustered [−0.4, +8.4]), widening to +8.4 at k=32 ([+6.1, +10.6]; clustered [+2.6, +14.7]). It reaches at least one correct sample for 47 of 49 conditions, against 39 for its parent. It does not restore breadth, though: distinct answers (1.87 vs. 1.82), entropy (0.28 vs. 0.30), and coverage (0.27 vs. 0.24) stay at the instruction-tuned level, far below the base model. AlphaMed also names off-list conditions more often than its parent, a mild loss of instruction following. By condition, AlphaMed slightly *raises* consideration in every severity tier (+2.2 to +4.3 points; Table 7). It still loses acute pulmonary edema (0.31 to 0.13), and it introduces an attractor of its own: spontaneous pneumothorax as the top answer in 10.3% of samples, against 4.3% for its parent and a reference rate of 1.0%. Medical post-training thus operates on an already collapsed distribution; the collapse comes from the general post-training stage. The other two descendants go further (Table 30). HuatuoGPT-o1 (SFT then PPO) and m1 (SFT distillation from a reasoning teacher) both *widen* the differential relative to their parent: 2.49 and 3.29 distinct answers vs. 1.82, and coverage@32 of 0.31 and 0.33 vs. 0.24. Their pass@32 rises by 13.7 [+11.3, +16.0] and 18.1 [+15.4, +20.8] points (clustered [8.0, 20.0] and [10.5, 25.9]). m1 gains almost nothing at k=1 (+0.9, interval includes zero), so its improvement lies entirely in breadth. Both reasoning-trained models are also less calibrated (ECE 0.17 and 0.21 vs. 0.11). In these three models, medical post-training on an instruction-tuned model does not add to the collapse, and reasoning distillation partially reverses it.

Medical RL from the base model. Alongside the instruct-initialized AlphaMed-7B, the AlphaMed authors released a companion checkpoint, AlphaMed-7B-base-rl, initialized from Qwen2.5-7B rather than Qwen2.5-7B-Instruct (according to its configuration file). The paper (Liu et al., 2025a) describes only the instruct-initialized models: GRPO with a binary correctness reward on about 19k multiple-choice questions from MedQA and MedMCQA, with a 64-question by 8-sample batch for 300 steps. It does not document the base-initialized checkpoint, so we cannot rule out differences in data or hyperparameters, and we treat it as corroborating rather than controlled evidence. It offers a third-party check on whether RL itself causes the collapse (Table 30, lower block). It does not. RL from the base model improves on the base at every k (+14.3 [+13.1, +15.7] points at k=1, +9.5 [+7.4, +11.6] at k=32). It keeps 3.14 distinct answers, against 1.87 for the instruct-initialized AlphaMed. Against that instruct-model AlphaMed, it is 2.2 [0.4, 3.9] points worse at k=1 and 10.8 [+8.6, +13.2] points better at k=32. Its severe-tier Considered@32 shows no detectable change relative to the base (−0.7 [−3.0, +1.7] points), against −12.4 for instruction tuning and −8.1 for instruction tuning followed by AlphaMed's RL (Table 7; all well below the chance levels of §D.2). RL from a base model also narrows less in OLMo's RL-Zero at T=1 and in our GRPO runs (§I), but not uniformly: RL-Zero at T=0.7 loses about as much as SFT, and within OLMo 3 7B's instruct pipeline the RL stage after SFT and DPO is the largest single narrowing (§D.4).

Model	Post-training	pass@1	pass@32	Distinct	Entropy	Cov@32	ECE
Qwen2.5-7B-Instruct	(parent)	0.478	0.598	1.82	0.30	0.239	**0.106**
AlphaMed-7B	rule-based RL	0.516	0.682	1.87	0.28	0.266	0.121
HuatuoGPT-o1-7B	SFT + PPO	**0.536**	0.735	2.49	0.44	0.306	0.168
m1-7B-23K	SFT distillation	0.486	0.779	3.29	0.62	0.328	0.206
Qwen2.5-7B (base)	(parent)	0.352	0.695	**4.91**	**1.00**	**0.335**	0.291
AlphaMed-7B (base-RL)	rule-based RL	0.495	**0.790**	3.14	0.62	0.320	0.202

Table 30: Medical models on DDXPlus (980 cases, 32 samples, T=0.7). Upper block: descendants of Qwen2.5-7B-Instruct; none narrows the differential further, and the reasoning-trained models widen it. Lower block: AlphaMed's companion checkpoint trained from the base model, which keeps most of the base model's diversity and has the best zero-shot pass@32 of any 7B model.

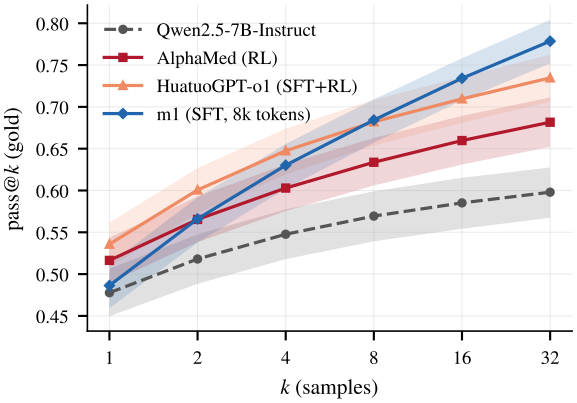


Figure 4: DDXPlus at T=0.7 (980 cases): Qwen2.5-7B-Instruct (dashed) vs. its medical descendants, with 95% case-bootstrap bands. Every descendant matches or beats its parent at every k, and the gaps widen with k: none of the three shows a sharpening tax at the medical stage.

H.2. Rare disease: overconfidence without accuracy

On 1,401 Phenopacket Store cases covering 749 diseases, every 7–8B model is near the floor: pass@1 is 1.4–2.2% and pass@32 3.7–7.0% for the Qwen-family models, and lower for OLMo 3 7B (2.5–4.5%). Stated confidence is not near the floor. Qwen2.5-7B-Instruct states a mean top-1 probability of 0.51, and its RL-trained medical descendants state 0.67 (HuatuoGPT-o1) and 0.69 (AlphaMed), while their accuracy is about 2%. ECE accordingly rises from 0.49 to 0.64–0.67; at this accuracy ECE mostly restates mean confidence, so we read it as a rise in confidence without a matching rise in accuracy. The RL-trained medical models (AlphaMed; HuatuoGPT-o1 after SFT) thus state much higher confidence where they are least accurate. The effect also appears without instruction tuning: AlphaMed's base-initialized checkpoint is the most overconfident Qwen-family model (mean confidence 0.71, ECE 0.70, accuracy 1.7%). The medical descendants do gain a little accuracy: Al-

Model	≤2022 (n=901) pass@1	pass@32	2023–24 (n=348) pass@1	pass@32	2025+ (n=152) pass@1	pass@32
Qwen2.5-7B (base)	2.0	8.2	0.0	0.3	0.6	3.3
Qwen2.5-7B-Instruct	2.3	7.7	0.0	0.9	0.8	3.9
AlphaMed-7B	3.0	9.0	0.3	1.4	1.4	3.9
AlphaMed-7B (base-RL)	2.4	8.8	0.1	1.1	1.4	4.6
HuatuoGPT-o1-7B	3.1	8.4	0.3	1.4	1.3	3.9
m1-7B	2.6	9.5	0.3	1.7	1.0	3.9
Qwen3-8B-Base	2.1	8.3	0.1	0.6	1.1	3.9
Qwen3-8B	3.1	5.2	0.1	0.3	2.0	2.6

Table 31: Phenopacket Store accuracy (%, exact diagnosis) by publication year of the source case report (1,401 cases, 32 samples, T=0.7). Every model is far more accurate on cases published up to 2022 than on 2023–24 cases; 2025+ cases fall in between.

phaMed beats its parent at every k (+0.6 [+0.2, +0.9] points at k=1, +1.0 [+0.1, +2.1] at k=32), HuatuoGPT-o1 at $k \leq 8$, and m1 at $k \geq 2$ (+1.4 [+0.4, +2.5] at k=32). These gains are an order of magnitude smaller than the rise in confidence. m1, trained only with supervised reasoning traces (at its intended 8k-token budget), is the exception on calibration and diversity: it is the most diverse model (15.3 distinct answers per 32 samples, against 11.5 for the parent and 9.2 for HuatuoGPT-o1), and its confidence rises least (0.57).

General post-training reproduces the DDXPlus pattern on Qwen3. Qwen3-8B beats its base at pass@1 (+0.7 [+0.4, +1.1]) and trails it from k=8 on, by −2.2 [−3.1, −1.3] points at pass@32 (a relative loss of 37%), with half the distinct answers (6.4 vs. 13.4). For Qwen2.5 no pass@k difference is significant. Unlike medical RL, general post-training *lowers* stated confidence here (0.61 to 0.46 for Qwen3, 0.61 to 0.51 for Qwen2.5), although base models also give invalid answers more often (8.5% of Qwen3-8B-Base samples). OLMo 3 does not share this: every stage of its pipeline states a mean confidence of 0.73–0.84 at about 1% accuracy (pass@32 2.5–4.5%), and its SFT checkpoint is the most overconfident model we evaluate (confidence 0.84, ECE 0.83). On the 500-case subset, our 8B GRPO runs (§I) are indistinguishable from their base (all pass@k differences within ±0.6 points).

Accuracy depends strongly on when the source case report was published (Table 31). For every model, pass@32 is 5–10% on cases published up to 2022 (n=901), at most 1.7% on 2023–24 cases (n=348), and 2.6–4.6% on 2025+ cases (n=152). The pattern is not monotone, since 2025+ cases score above 2023–24 cases, so case mix also matters; the gap between older and recent cases is consistent with, but does not establish, memorization of older case reports.

H.3. Published case reports

MedCaseReasoning presentations stop before the authors' differential, so these cases test open-ended diagnosis from real presentations (prompts and grading in Appendix B). We sampled the four zero-shot Qwen and OLMo pairs; OLMo 3.1 32B was left out for cost and Mistral because its base model needs worked examples, which this dataset cannot supply without reference lists. The two judges agree on 97.6% of 391,987 distinct names

(κ 0.77; with only 4.5–6.3% of names accepted, most agreement is on rejections); Phi-4 accepts more names, and the strict grading leaves every direction unchanged. The authors' alternatives were extracted by a language model from the reports' reasoning notes; some are broad ("bacterial", "viral"), which only makes them easier to match. For voting, agreement and distinct answers, top answers are grouped by their normalized text whether correct or not; an earlier version of this analysis merged all correct answers into one group, which inflated voting gains and agreement for every model. Against the alternatives the authors wrote down, the stated/sampled split replicates (Table 32): post-trained samples cover 8–14 points fewer of them for the Qwen pairs (-3.0, n.s., for OLMo 3 7B), one list covers them as before, and for Qwen3-8B and Qwen2.5-32B one list covers more of them than 32 samples, whereas base models' samples cover more than their lists (by 0.5–10.0 points, n.s. for Qwen2.5-32B). The consequences follow DDXPlus in direction but are smaller. Voting helps little for any model ($+1.6$ to $+2.7$ points for base models, -0.7 to $+3.0$ after post-training), partly because paraphrases of the same diagnosis count as different answers. After post-training the stated probability ranks correctness at least as well as agreement (Qwen3-8B $+0.085$ $[0.020, 0.146]$, Qwen2.5-32B $+0.062$ $[0.006, 0.120]$, the others n.s.), and it gives higher accuracy at 50% coverage for all four post-trained models; for base models agreement is as good or, for OLMo 3 7B, better (-0.172). OLMo-3-7B-Instruct gives more distinct answers than its base model (24.6 against 13.7 among valid samples), whose answers are valid only 67% of the time. Accuracy does not differ between cases published up to 2022 and from 2023 (e.g. Qwen2.5-32B-Instruct pass@32 0.49 and 0.52); since both periods may precede the models' training cutoffs, this does not rule out memorization.

H.4. Stated and sampled readouts on real rare-disease cases

Phenopacket Store cases are real, but they have no reference differential and accuracy is near the floor, so we can test only the consequences of §4.3, and we count an answer as correct if it is the diagnosis or a closely related Mondo class (exact matches give only 8–31 correct single-call answers per model). Top answers are grouped by Mondo class for voting and agreement (Table 33). The DDXPlus shifts reappear, but the top-three comparison does not flip as it does on DDXPlus. Majority voting over 32 samples helps the base models by 3.1–4.6 points but the final post-trained models by only 0.4–1.5 (Qwen3-8B $+0.4$, n.s.). Post-training weakens agreement as an error signal (Qwen3-8B 0.830 $\rightarrow$ 0.763, Qwen2.5-7B 0.845 $\rightarrow$ 0.745, OLMo 3 7B 0.814 $\rightarrow$ 0.726) and strengthens the stated probability (0.567 $\rightarrow$ 0.707, 0.598 $\rightarrow$ 0.717, 0.464 $\rightarrow$ 0.652). Agreement beats the stated probability by 0.25–0.35 AUROC for the base models and by only 0.03–0.07 for the post-trained ones, significantly only for Qwen3-8B (-0.056 $[-0.120, -0.002]$). On these open-ended cases the three most frequent of 32 sampled answers still contain a correct diagnosis more often than one list's top three, but the gap shrinks from 4.6–7.0 points for the base models to 2.7–3.6 for Qwen2.5-7B-Instruct and OLMo-3-7B-Instruct and to a tie for Qwen3-8B ($+0.4$ $[-0.8, +1.8]$). The medical models behave like their post-trained parent, with agreement the better signal for all of them.

Pair	Model	Valid	pass@1	pass@32	Distinct	Recall@5 one list	Authors' alternatives one list	Authors' alternatives 32 samples	Vote@32 − single	AUROC, single-call answer correct stated p	AUROC, single-call answer correct agree@32	AUROC, single-call answer correct stated − agree
Qwen3-8B	base	0.878	0.139	0.433	13.02	0.301	0.303	0.326	+1.8	0.558	0.602	$-0.044^{\dagger}$
	post-trained	1.000	0.180	0.308	6.63	0.352	0.296	0.186	+0.8	0.678	0.592	+0.085
Qwen2.5-7B	base	0.986	0.134	0.438	15.75	0.263	0.290	0.378	+1.6	0.565	0.594	$-0.029^{\dagger}$
	post-trained	1.000	0.154	0.407	13.73	0.293	0.285	0.289	+0.8	0.623	0.593	$+0.030^{\dagger}$
Qwen2.5-32B	base	0.985	0.206	0.530	13.77	0.371	0.322	0.326	+2.7	0.666	0.621	$+0.045^{\dagger}$
	post-trained	1.000	0.251	0.495	11.51	0.429	0.320	0.245	-0.7	0.650	0.588	+0.062
OLMo 3 7B	base	0.669	0.073	0.390	13.70	0.221	0.251	0.349	+2.2	0.462	0.635	−0.172
	post-trained	0.994	0.120	0.370	24.61	0.240	0.237	0.319	+3.0	0.675	0.653	$+0.023^{\dagger}$

Pair	Post-trained − base (primary grading) pass@1	pass@32	Distinct	recall@5	alt., list	alt., samples	List − samples (post, alt.)	Vote − single (post)	Strict grading pass@32	Strict grading recall@5
Qwen3-8B	+4.1	−12.5	−6.39	+5.1	$-0.7^{\dagger}$	−14.0	+11.0	$+0.8^{\dagger}$	−12.2	+5.0
Qwen2.5-7B	+2.0	$-3.1^{\dagger}$	−2.02	+3.0	$-0.5^{\dagger}$	−8.9	$-0.4^{\dagger}$	$+0.8^{\dagger}$	$-3.2^{\dagger}$	+2.8
Qwen2.5-32B	+4.4	$-3.5^{\dagger}$	−2.27	+5.8	$-0.2^{\dagger}$	−8.1	+7.5	$-0.7^{\dagger}$	$-3.0^{\dagger}$	+5.7
OLMo 3 7B	+4.7	$-2.0^{\dagger}$	+10.91	$+1.9^{\dagger}$	−1.4	$-3.0^{\dagger}$	−8.2	+3.0	$-1.6^{\dagger}$	$+1.8^{\dagger}$

Table 32: Published case reports (MedCaseReasoning test split, 897 PMC case reports, zero-shot, 32 samples, T=0.7). A name is correct if the primary judge (Mistral-Small-24B-Instruct) says it names the report's final diagnosis; *strict*: only if Phi-4 agrees as well (the judges agree on 97.6% of 391,987 judged names, Cohen's κ 0.77). *Authors' alternatives*: share of the diagnoses other than the final one that the report's authors considered (extracted from its reasoning notes; 893 cases with at least one, 3.5 on average) that one list names, or that any of the 32 sampled top answers names; *list − samples*: the difference within the post-trained model. Invalid samples count as wrong for pass@k; the list and agreement columns use valid samples. Distinct: distinct top answers among valid samples, grouped by normalized text whether correct or not (as for voting and agreement). Differences in points. †: not significant (paired case bootstrap, 10,000 resamples, Holm over the 36 pair tests; for the AUROC and list − samples differences, the 95% interval includes zero).

Appendix I. GRPO with a group-level reward

We also fine-tuned Qwen3-1.7B (instruct) and Qwen3-8B-Base with GRPO (LoRA, 200 steps, 16 samples per prompt, KL coefficient 0.04) on DDXPlus training patients disjoint from the test split, comparing the standard correctness reward (`acc`) with a group-level reward (`gm`). If k_a of the n samples in a GRPO group give top answer a, `gm` adds $\mathrm{clip}(n\,p_{\mathrm{ref}}(a) - (k_a - 1), 0, 1)$, which pays an answer only while its condition is under-represented relative to its reference mass. At 1.7B, a boosted variant (weight 3, training temperature 1.2, so the two changes are confounded) raised pass@1 by 1.6 and distinct answers by 0.10 over `acc` (two seeds; intervals ignore seed variation). From Qwen3-8B-Base, both recipes reached nearly the official Qwen3-8B's top-1 accuracy (0.537 and 0.539 vs. 0.558) while keeping more than twice its distinct answers (2.7–2.9 vs. 1.27), and `gm` narrowed the base model less than `acc` (distinct answers +0.14 at equal pass@1). Neither changed Phenopacket accuracy. Because `gm` needs a reference differential at training time, which real clinical data rarely provide, and the runs train on the evaluation distribution, these are existence results only. Closely related rewards computed from group answer counts have appeared since: a witness advantage that trains models to sample the distributions they

Model	Single-call correct	Vote@32 gain	Top-3: list − 32 samples	AUROC, single-call answer correct			Correct answers
				Stated p	Agree@5	Agree@32	
Qwen3-8B-Base	0.074	+4.6	−7.0	0.567	0.768	**0.830**	104
Qwen3-8B	0.080	+0.4	+0.4	0.707	0.726	**0.763**	112
Qwen2.5-7B	0.062	+3.8	−5.4	0.598	0.788	**0.845**	87
Qwen2.5-7B-Instruct	0.056	+1.5	−3.6	0.717	0.661	**0.745**	78
OLMo-3-1025-7B	0.060	+3.1	−4.6	0.464	0.765	**0.814**	84
SFT	0.035	+1.1	−3.6	0.628	0.730	**0.814**	49
SFT+DPO	0.026	+1.2	−3.3	0.651	0.673	**0.725**	36
SFT+DPO+RLVR (Instruct)	0.035	+0.9	−2.7	0.652	0.644	**0.726**	49
RL-Zero (from base)	0.065	+2.3	−3.1	0.583	0.791	**0.827**	91
AlphaMed-7B	0.071	+1.6	−3.0	0.586	0.755	**0.797**	100
HuatuoGPT-o1-7B	0.093	+2.6	−5.7	0.653	0.708	**0.756**	130
m1-7B	0.084	+2.9	−7.9	0.650	0.747	**0.800**	117
AlphaMed-7B (base-RL)	0.054	+2.5	−4.2	0.672	0.733	**0.789**	75

Table 33: Aggregation and uncertainty on Phenopacket Store (1,401 real rare-disease cases, 32 samples, T=0.7). An answer counts as correct if it is the diagnosis or a closely related Mondo class. Columns as in Table 29; Correct answers: number of correct single-call answers, the positives for the AUROCs. Bold: best signal.

state (Arabi et al., 2026), and a batch bonus that penalizes repeated answers (Song et al., 2025); we do not compare against them. The GRPO runs enter only the 56-combination audit (§G.3).

Appendix J. Condition-clustered intervals and multiple comparisons

DDXPlus has 20 test cases for each of 49 true pathologies, and effects can be specific to a condition, so cases are not independent. Table 34 gives the headline base-vs-post differences with a bootstrap that resamples the 49 pathologies with all their cases (50,000 resamples for these headline tests, so that results near the threshold do not flip between runs; 10,000 elsewhere). Two-sided p-values use the add-one correction, $2(1 + \min(n_{\leq 0}, n_{\geq 0}))/(B + 1)$, so the smallest attainable value lies below the strictest Holm threshold, 0.05/112. Holm correction runs over the 112 tests (8 pairs × 14 metrics fixed in advance: pass@1, pass@32, distinct answers, coverage@32, sampled TV, list TV, mass named, severe named, list and sample TV without the true pathology, DDF1, unwarranted severe conditions named, and the two AUROCs; the table shows seven). The same procedure is applied separately to the few-shot pairs (70 tests), the Mistral and thinking-mode pairs (42) and the stage and medical comparisons (154). Not surviving correction among the headline tests shown, marked †: list TV without the true pathology for Qwen3-8B at T=0.7; sampled TV for Qwen2.5-7B at T=0.7; sampled TV for Qwen2.5-7B at T=1; pass@32 for Qwen2.5-32B at T=0.7; list TV for OLMo 3 7B at T=0.7; DDF1 for OLMo 3 7B at T=0.7; agreement AUROC for OLMo 3 7B at T=0.7; list TV for OLMo 3 7B at T=1; list TV without the true pathology for OLMo 3 7B at T=1; agreement AUROC for OLMo 3 7B at T=1; pass@32 for OLMo 3.1 32B at T=0.7; sampled TV for OLMo 3.1 32B at T=0.7; agreement AUROC for OLMo 3.1 32B at T=0.7. OLMo 3 7B's list TV without the true pathology at T=0.7 survives as a significant *increase*.

Pair (post − base)	pass@32 (pts)	Distinct	TV samples	TV one list	TV list w/o gold	DDF1	AUROC agree@32
Qwen3-8B, T=0.7	−12.4	−2.50	+0.043	−0.043	−0.018†	+0.042	−0.160
	[−19.6, −6.0]	[−2.89, −2.13]	[+0.021, +0.064]	[−0.059, −0.028]	[−0.032, −0.005]	[+0.029, +0.056]	[−0.216, −0.096]
Qwen3-8B, T=1	−14.6	−5.06	+0.067	−0.074	−0.039	+0.066	−0.158
	[−21.5, −8.3]	[−5.49, −4.65]	[+0.042, +0.091]	[−0.090, −0.059]	[−0.052, −0.027]	[+0.053, +0.080]	[−0.211, −0.104]
Qwen2.5-7B,	−9.7	−3.09	+0.028†	−0.062	−0.041	+0.062	−0.106
T=0.7	[−16.1, −4.1]	[−3.63, −2.57]	[+0.007, +0.049]	[−0.076, −0.049]	[−0.054, −0.028]	[+0.049, +0.075]	[−0.163, −0.044]
Qwen2.5-7B, T=1	−10.3	−5.59	+0.042†	−0.088	−0.056	+0.083	−0.104
	[−17.2, −4.2]	[−6.03, −5.15]	[+0.015, +0.068]	[−0.103, −0.074]	[−0.069, −0.043]	[+0.069, +0.097]	[−0.159, −0.046]
Qwen2.5-32B,	−4.9†	−2.10	+0.033	−0.053	−0.043	+0.051	−0.079
T=0.7	[−10.3, +0.5]	[−2.49, −1.74]	[+0.018, +0.049]	[−0.064, −0.042]	[−0.053, −0.032]	[+0.041, +0.061]	[−0.119, −0.038]
OLMo 3 7B,	−28.3	−2.56	+0.090	+0.029†	+0.034	−0.006†	−0.076†
T=0.7	[−35.6, −21.1]	[−2.84, −2.27]	[+0.071, +0.109]	[+0.004, +0.052]	[+0.017, +0.051]	[−0.022, +0.009]	[−0.157, +0.001]
OLMo 3 7B, T=1	−15.2	−3.95	+0.079	−0.009†	+0.006†	+0.024	−0.076†
	[−20.8, −9.7]	[−4.22, −3.68]	[+0.060, +0.098]	[−0.031, +0.011]	[−0.006, +0.018]	[+0.010, +0.036]	[−0.151, −0.011]
OLMo 3.1 32B,	−4.0†	−2.18	+0.020†	−0.045	−0.023	+0.040	−0.056†
T=0.7	[−6.8, −1.2]	[−2.47, −1.90]	[+0.002, +0.037]	[−0.059, −0.031]	[−0.034, −0.012]	[+0.026, +0.054]	[−0.095, −0.019]

Table 34: Base → post-trained (final Instruct) differences on DDXPlus with 95% intervals from a bootstrap over the 49 true-pathology clusters. †: not significant after Holm correction across 112 tests (50,000 resamples, add-one p-values). pass@32 here is recomputed from the samples with the analysis scripts' name matching and can differ from Table 6 by up to 0.3 points; the AUROC differences come from the same script and differ from those implied by Table 29 by up to 0.006.

Pooled effects and equivalence. Statements such as "robust only for Qwen" compare significance across pairs. Table 35 instead pools each difference across pairs with a joint bootstrap over the 49 pathologies, which keeps the correlation between pairs scored on the same cases, and tests whether Qwen differs from the other families. The sampled collapse, the stated-list gain in recall and the weaker agreement are significant when pooled in all three prompt settings; agreement's AUROC falls by 0.10 zero-shot and 0.18–0.19 with worked examples. The stated list gains more for Qwen than for the other families in TV, conditional TV and severe conditions named in every setting, and in recall zero-shot and with reference examples; the zero-shot pass@32 loss is smaller for Qwen (by 7.1 points), mostly because OLMo 3 7B loses 28; and agreement falls more for Qwen only with reference examples. For claims of no difference we use two one-sided tests with margins fixed in advance (5 points for accuracy and recall, 0.02 for TV, 0.03 for AUROC). Thinking mode's pass@32 is at least its base model's (90% interval $[+2.6, +14.1]$ points), but the non-significant zero-shot pass@32 losses at 32B and OLMo 3 7B's recall change ($+1.5$, $[-2.9, +5.9]$) are inconclusive rather than equivalent to zero.

Post-trained − base	Zero-shot (5 pairs) pooled	Zero-shot (5 pairs) Qwen − other	Worked examples (6) pooled	Worked examples (6) Qwen − other	Another model's examples (6) pooled	Another model's examples (6) Qwen − other
pass@32 (points)	−11.9 [-15.4, -8.5]	+7.1 [+2.6, +11.6]	−19.7 [-25.7, -14.1]	+2.9 [-2.8, +8.1]†	−18.1 [-23.8, -12.5]	+3.7 [-1.1, +8.2]†
distinct answers	−2.49 [-2.78, -2.20]	−0.20 [-0.53, +0.09]†	−3.86 [-4.47, -3.24]	+1.05 [+0.69, +1.41]	−3.86 [-4.44, -3.27]	+2.34 [+1.99, +2.71]
recall@5, one list (points)	+11.2 [+8.1, +14.4]	+6.8 [+3.6, +10.0]	+13.9 [+8.8, +18.9]	+4.9 [+0.1, +10.0]	+13.5 [+9.3, +17.8]	+3.6 [-1.1, +8.3]†
TV, one list	−0.035 [-0.045, -0.025]	−0.044 [-0.058, -0.031]	−0.044 [-0.058, -0.031]	−0.063 [-0.076, -0.050]	−0.048 [-0.060, -0.036]	−0.041 [-0.055, -0.029]
conditional TV	−0.031 [-0.043, -0.020]	−0.064 [-0.092, -0.037]	+0.004 [-0.018, +0.027]†	−0.052 [-0.078, -0.028]	−0.057 [-0.073, -0.040]	−0.034 [-0.055, -0.014]
severe named (points)	+7.2 [+4.3, +10.4]	+5.1 [+2.3, +7.8]	+5.4 [+1.6, +9.3]	+8.3 [+4.7, +11.7]	+4.9 [+1.7, +8.3]	+7.0 [+3.1, +11.0]
AUROC, agreement	−0.095 [-0.125, -0.061]	−0.049 [-0.098, +0.005]†	−0.190 [-0.229, -0.146]	−0.058 [-0.100, -0.010]	−0.182 [-0.225, -0.130]	−0.002 [-0.039, +0.037]†

Table 35: Pooled base → post-trained effects across pairs (DDXPlus, T=0.7). *Pooled*: mean of the pairs' differences; *Qwen − other*: mean over the Qwen pairs minus the mean over the OLMo (and Mistral) pairs. 95% intervals come from a joint bootstrap that draws the 49 pathologies once per resample and applies the draw to every pair, so the correlation between pairs scored on the same cases is kept (2,000 resamples). †: interval includes zero.